İSAK

# MILLIY SERTIFIKAT ORIGINAL SAVOLLARI

(MAVZULASHTIRILGAN)

- MILLIY SERTIFIKAT TESTLARI
- TESTLARNING VIDEOYECHIMLARI

+998 99 340 06 90

Bilim va malakalarni baholash agentligi

# BILIM VA MALAKALARNI BAHOALSH AGENTLIGI

*Bilimingga ishon va muvaffaqiyatga erish!*

## UMUMTA'LIM VA MAXSUS FANLARDAN BILISH DARAJASINI ANIQLASHNING MILLIY SERTIFIKAT TEST IMTIHONI ISHTIROKCHILARI UCHUN

# FIZIKA

## FANIDAN

## TEST TOPSHIRIQLARI KITOBI

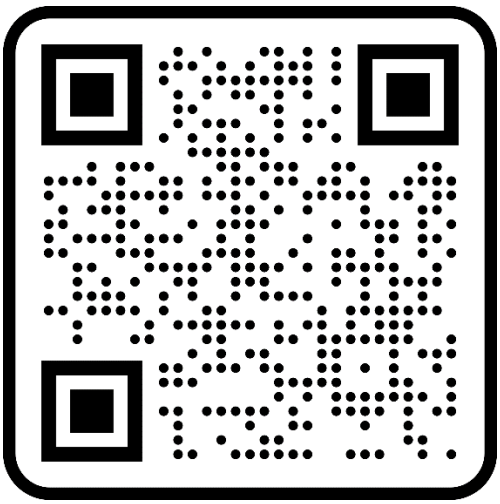
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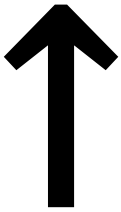
*Talabgorning familiyasi, ismi va otasining ismi* *Imzo*

**Testni tuzuvchi:** *Abdumajidov Shahzodbek Rahmiddin o'g'li*

**Test natijalari quyidagi telegram kanalimizda e'lon qilinadi:** *https://t.me/isak_team*

Ushbu test varianti 45 ta (1—45) topshiriqdan iborat.
Test topshiriqlari **RASH METODI** asosida baholanadi. Test topshiriqlari kitobida keltirilgan ball xom ball hisoblanadi.
Kitobda yopiq va ochiq turdagi test topshiriqlari mavjud:
— yopiq turdagi test topshiriqlarida bitta javobni (A, B, C yoki D) tanlang va javoblar varaqasidagi topshiriq raqamiga mos qatorga yozing;
— ochiq turdagi test topshiriqlarining javobini javoblar varaqasidagi topshiriq raqamiga mos qatorga aniq, tushunarli va eng sodda holatda yozing;
— moslashtirishni talab qiluvchi yopiq test topshiriqlari uchun umumiy oltita (A—F) javob varianti berilgan, uchta topshiriqqa (33-, 34-, 35-test topshiriqlariga) ushbu javoblar orasidan mos ravishda bittadan javob tanlang va javoblar varaqasiga belgilang.



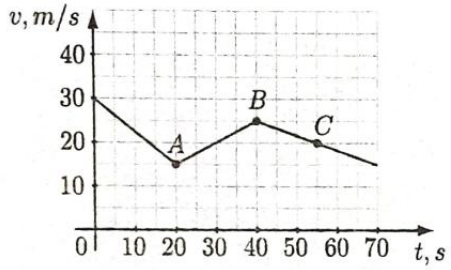


[Kitobdagi har bir masalaning yechimi ushbu youtube sahifamizda mavjud.]

## MEXANIKA
## KINEMATIKA

1. Avtomobilning boshlang‘ich tezligi 10 $m/s$, tezlanishi $-2\ m/s^2$ ga teng. U qancha $(m)$ masofa bosib o‘tib to‘xtashini aniqlang.
A) 25
B) 100
C) 20
D) 50

2. Grafikda to‘g‘ri chiziqli harakatlanayotgan jism tezligining o‘zgarishi tasvirlangan. Grafikning $BC$ qismida jism bosib o‘tgan masofani $(m)$ toping.




A) 475
B) 337,5
C) 425,5
D) 400

3. Harakat tenglamasi $x = 16 - 8t^2\ [m]$ bo‘lgan jismning dastlabki sekunddagi ko‘chishini toping $(m)$.
A) 2
B) 4
C) 8
D) 6

4. Pastga erkin tashlangan jism birinchi sekundda 12,6 $m$ yo‘l bosdi. Shu jism ikkinchi sekundda qancha yo‘l $(m)$ yuradi?
A) 22,6
B) 18,4
C) 27,6
D) 20,8

5. Jism tik pastga tashlandi. Agar u birinchi sekundda 15 $m$ ga pastlagan bo‘lsa, uchinchi sekund oxiridagi tezligini $(m/s)$ toping.
A) 60
B) 30
C) 50
D) 40

6. Jism $x = 14 - 8t^2\ [m]$ qonuniyat bo‘yicha harakatlanmoqda. Jismning dastlabki 1 $s$ da bosib o‘tgan yo‘lini $(m)$ toping

A) 4
B) 8
C) 10
D) 6

7. Oyda 20 $m/s$ tezlik bilan 30° burchak ostida jism otildi. Jismning parvoz vaqti $(s)$ qancha? $g_{oy} = 5/3\ m/s^2$.
A) 15
B) 4
C) 12
D) 30

8. Biror burchak tezlanish bilan aylanayotgan disk 6 $s$ da 4 marta aylandi. Uning aylanish oxiridagi burchak tezligi $(rad/s)$ qanday?
A) $4\pi/3$
B) $8\pi/3$
C) $5\pi/3$
D) $2\pi$

9. Vertikal yuqoriga 50 $m/s$ tezlik ilan otilgan jismning uchish vaqtini $(s)$ toping.
A) 10
B) 20
C) 12
D) 6

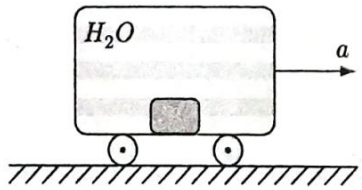
10. Tezligi 6 $m/s$ bo‘lgan jism 2 $s$ vaqt davomida qanday masofa $(m)$ bosib o‘tadi?
A) 12
B) 6
C) 4
D) 16

11. Kosmosda birin-ketin (bir xil boshlang‘ich tezlik bilan) otilgan jismlarning nisbiy tezligi qanday o‘zgaradi? O‘zaro ta’sirdan boshqa kuch mavjud emas.
A) Notekis kamayadi
B) Notekis ortadi
C) O‘zgarmaydi
D) Tekis kamayadi

12. 30 $m/s$ tezlik bilan harakatlanayotgan avtomobil 18 $km$ masofani necha minutda bosib o‘tadi?
A) 5
B) 10
C) 600
D) 300

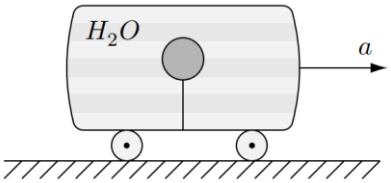
13. Rasmda ko‘rsatilgan suv bilan to‘ldirilgan sisterna o‘ng tomonga 4 $m/s^2$ tezlanish bilan harakat

qilmoqda. Sisterna tubida yotgan alyuminiy bo‘lakchasining yerga nisbatan tezlanishi ($m/s^2$) qanchaga teng?




A) 3
B) 2,5
C) 1,5
D) 4

14. Rasmda ko‘rsatilgan sisterna o‘ng tomonga tezlanish bilan harakat qilsa, sisterna ichidagi plastik idish qaysi tomonga og‘adi?




A) tezlanish yo‘nalishida
B) tezlanish yo‘nalishiga qarshi
C) o‘quvchi tomon yo‘nalgan
D) qog‘oz tekisligidan orqaga

15. Bolalar lagerdan qaytishda avtobus 70 $km/h$ o‘zgarmas tezlik bilan harakat qilmoqda. Yo‘lda havo o‘zgarib yomg‘ir yog‘ishni boshladi va avtobus 50 $km/h$ tezlik bilan harakatini davom ettirdi. Shunda avtobus manzilga belgilangan vaqtdan 10 $minut$ kechikib yetib kelgan bo‘lsa, qancha vaqt ($min$) sekinlashgan tezlik bilan yurgan?
A) 15
B) 35
C) 21
D) 10

16. Jismlar ikki xil tezlanishda harakatlanmoqda: $a_1 = 5\ m/s^2$ va $a_2 = 20\ m/min^2$. Bu tezlanishlarni o‘zaro taqqoslang.
A) $a_1 < a_2$
B) $a_1 > a_2$
C) $a_1 = a_2$
D) taqqoslab bo‘lmaydi

17. Birinchi kema 4 $uzel$ tezlik bilan g‘arbga, ikkinchi kema esa 6 $uzel$ tezlik bilan aniq shimoli-sharqqa harakatlanmoqda. Ularning nisbiy tezligini uzelda toping. $Uzel$ - dengiz kemalarining tezligi o‘lchanadigan birlik.
A) 9,3
B) 7,4

C) 8,5
D) 7,9

18. Agar $a$ – umumiy tezlanish, $a_n$ – normal tezlanish, $a_t$ – tangensial tezlanish bo‘lsa, quyidagilardan qaysi birlari o‘rinli bo‘lishi mumkin?
1) $a_n > a$ 2) $a \geq a_t$ 3) $a_t > a$ 4) $a \geq a_n$
A) 1, 3
B) 2, 4
C) 1, 2, 4
D) faqat 4

19. Jism boshlang‘ich tezliksiz 5 $s$ davomida 7,5 $m$ masofa bosdi. Keyin 12 $s$ davomida o‘zgarmas tezlik bilan harakat qildi. So‘ngra 6 $s$ da esa tekis sekinlashib to‘xtadi. Jismning oxirgi 12 $s$ ichidagi o‘rtacha tezligini $(m/s)$ aniqlang.
A) 2,25
B) 2,5
C) 4,5
D) 9

20. Agar $a$ - umumiy tezlanish, $a_n$ - normal tezlanish, $a_t$ - tangensial tezlanish bo‘lsa quyidagilardan qaysilari to‘g‘ri?
1 ) $a_n > a$ 2) $a_t > a$ 3) $a \geq a_t$ 4) $a \geq a_n$
A) 1, 3
B) 2, 4
C) 3, 4
D) 1, 2

21. Agar 1-jism sharq 5/13 m/s tezlik bilan ikkinchi jism shimolga 12/13 m/s tezlik bilan harakatlanayotgan bo‘lsa ularning nisbiy tezligi qanday?
A) 1
B) 7/13
C) 17/13
D) 12/5

22. Jismning harakat tenglamasi $x = -10 + 12t - 2t^2$ $[m]$ bilan o‘zgarmoqda. Jismni to‘xtaguncha bosib o‘tgan yo‘lini $(m)$ toping.
A) 72
B) 144
C) 18
D) 36

ISAK

## DINAMIKA

1. Silliq gorizontal sirtda turgan 5 $kg$ massali jismga 20 $N$ va 55 $N$ kuchlar ta’sir qilmoqda. Jisnming minimal tezlanishini $(m/s^2)$ toping.

A) 4
B) 7
C) 11
D) 15

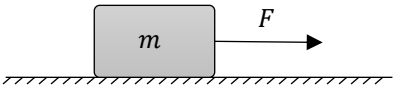
2. Tormoz bergan avtomobil asfaltda 25 $m$ iz qoldirib to‘xtagan. Agar g‘ildiraklar bilan asfalt orasidagi ishqalanish koeffitsiyenti 0,45 bo‘lsa, avtomobilning dastlabki tezligini $(m/s)$ toping.

A) 18
B) 17
C) 16
D) 15

3. Massasi $m_1$ bo‘lgan jismni Yer 680 $N$ kuch bilan tortadi. Massasi $m_2$ bo‘lgan jismni Oy 960 $N$ kuch bilan tortadi. Agar $g_{oy} = \frac{g_{yer}}{6}$ bo‘lsa, $\frac{m_1+m_2}{m_1}$ nisbatni aniqlang.

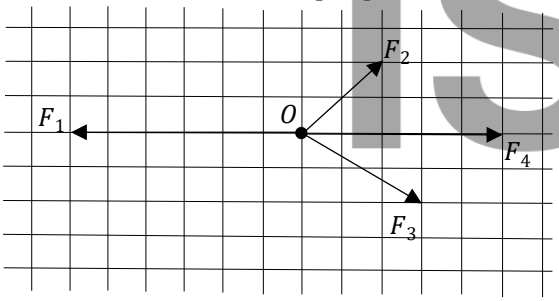
A) 161/17
B) 121/17
C) 161/13
D) 121/13

4. Gorizontal sirtda turgan massasi 10 $kg$ bo‘lgan jismga rasmda ko‘rsatilgandek kuch ta’sir etmoqda. Birinchi holatda jism 60 $N$ kuch bilan, so‘ngra 100 $N$ kuch bilan tortildi. Agar jism va sirt orasidagi ishqalanish koeffitsiyenti 0,8 ga teng bo‘lsa, ikkala holdagi ishqalanish kuchlarining nisbatini toping.




A) 1/3
B) 3/4
C) 5/3
D) 5/2

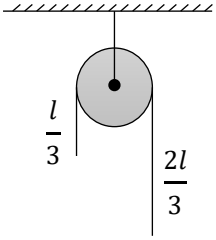
5. Rasmda $O$ nuqtaga ta’sir etuvchi to‘rtta kuch tasvirlangan. Agar $F_1 = 12\ N$ va $F_4 = 10\ N$ bo‘lsa, teng ta’sir etuvchi kuchni toping.




A) 2
B) 20

C) 15
D) 8

6. Massasi $m$ bo‘lgan arqon rasmda ko‘rsatilgan holatda bo‘lganda uning blok o‘qiga ta’sir qiluvchi kuchini toping. $g$ — erkin tushish tezlanishi.



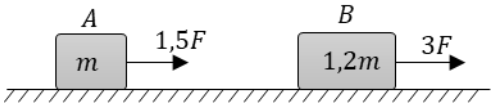

A) $\frac{3}{4}mg$
B) $\frac{5}{3}mg$
C) $\frac{3}{5}mg$
D) $\frac{8}{9}mg$

7. Jismga og‘irlik kuchi 6 $N$, sharqqa 9 $N$, shimolga 6 $N$ va janubga 4 $N$ kuch ta’sir etmoqda. Jismga ta’sir etayotgan kuchlarning teng ta’sir etuvchisini toping($N$).
A) 11
B) 8
C) 2
D) 5

8. Birinchi jismning oydagi og‘irligi 60 $N$ ga teng. Ikkinchi jismning yerdagi og‘irligi 90 $N$ ga teng. Agar $g_{oy} = \frac{g_{yer}}{6}$ bo‘lsa, $\frac{m_1+m_2}{m_2}$ nisbatni aniqlang.
A) 6
B) 10
C) 5
D) 2

9. 25 $g$ jismning og‘irlik kuchi qancha ($N$)?
A) 0,025
B) 0,25
C) 0,0025
D) 2,5

10. Rasmda $A$ va $B$ jismlar va ularga ta’sir etadigan kuchlar tasvirlangan. Agar $B$ jism 4 $s$ vaqt davomida 84 $m$ yo‘l bosib o‘tgan bo‘lsa, $A$ jismning 6 $s$ da bosib o‘tgan yo‘lini aniqlang ($m$).



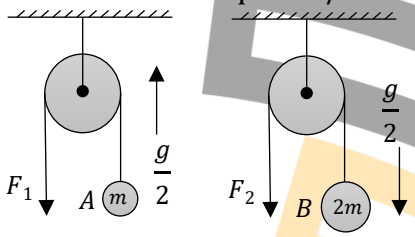

A) 120
B) 84,4
C) 113,4

D) 90

11. Massasi 2 $kg$ bo‘lgan jism ishqalanish koeffitsiyenti 0,8 bo‘lgan aravacha ustida tinch turibdi. Agar aravacha 7,5 $m/s^2$ tezlanish bilan harakatlansa, jismga ta’sir qiluvchi ishqalanish kuchini ($N$) toping.
A) 16
B) 15
C) 10
D) 20

12. 1,2 $m$ uzunlikdagi chilvirga bog‘langan 0,9 $kg$ massali shar vertikal tekislikda aylantirilmoqda. Pastki nuqtada sharning tezligi 4 $m/s$ bo‘lsa, shu paytda chilvirning tarangligini ($N$) toping.
A) 3
B) 9
C) 21
D) 19

13. Rasmda tasvirlangan vaznsiz bloklarga $m$ va $2m$ massali sharchalar osilgan. $A$ sharcha $g/2$ tezlanish bilan vertikal yuqoriga, B sharcha esa $g/2$ tezlanish bilan vertikal pastga harakatlantirilmoqda. $F_1/F_2$ nisbatni toping.

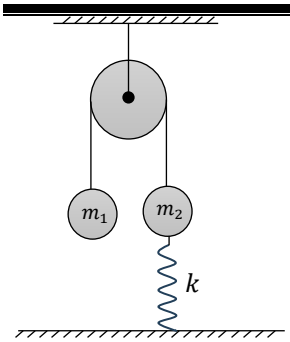



A)1
B) 1/3
C) 3
D) 3/2

14. Qo‘zg‘almas blokka 7 $kg$ va 4 $kg$ massali jismlar osilgan. 7 $kg$ massali jismni 3 $m$ balandlikdan quyib yuborilsa, u yerga tekkan vaqtda 4 $kg$ massali jismning tezligi qanday bo‘ladi($m/s$)?
A) 16
B) 4
C) 32
D) 2

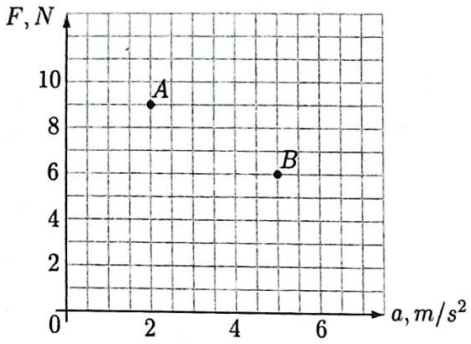
15. Yer sirtida birinchi kosmik tezlik 8 $km/s$ ga teng. Agar Yer radiusi 6400 $km$ ga teng bo‘lsa, Yer sirtidan qanday balandlikda ($km$) birinchi kosmik tezlik $5\sqrt{2}$ $km/s$ ga teng bo‘ladi?
A) 1792
B) 1297
C) 1600
D) 1486

16. Rasmdagi sistema muvozanatda turibdi. Bunda: $m_1$=8 $kg$, $m_2$=3 $kg$ va $k$=500 $N/m$. Prujina qanchaga cho‘zilganini aniqlang($cm$).




A) 6
B) 5
C) 8
D) 10

17. Quyida jismlarga ta'sir etuvchi natijaviy kuchning ularning tezlanishiga bog'lanish grafigi keltirilgan.




Grafikka asosan $A$ jismning $B$ jism massasiga nisbatini ( $\frac{m_A}{m_B}$ ) aniqlang.

A) 5/3
B) 5/4
C) 15/4
D) 5/2

18. Bikirligi 1,6 $kN/m$ bo'lgan prujinaga 5 $kg$ massali yuk osilgan. Prujina qanchaga $(cm)$ cho'ziladi?
A) 25/8
B) 5/8
C) 50/8
D) 500/8

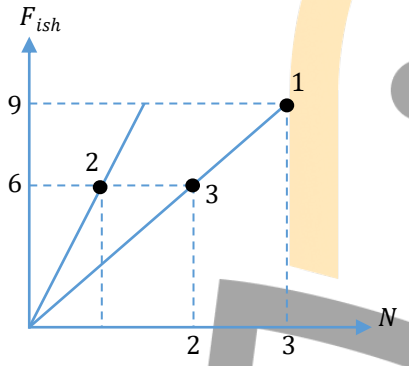
19. Massasi 300 $g$ bo'lgan jism 2 $m/s^2$ tezlanish bilan harakatlanmoqda. Natijaviy kuchni $(N)$ aniqlang.
A) 0,6
B) 1,2
C) 1,8
D) 0,9

20. Massasi 2,5 $kg$ bo'lgan jism qiyalik burchagi 30° bo'lgan va qiya tekislikka parallel 12 $N$ kuch bilan ushlab turilibdi. Jism va sirt orasidagi ishqalanish kuchini $(N)$ toping.
A) 0,5

B) 25
C) 12
D) 12,5

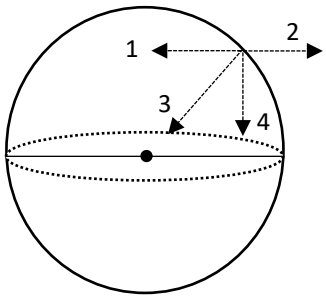
21. Prujinaga 50 $g$ yuk osilganda 16 $sm$ ga cho‘zilgan bo‘lsa prujinaning bikirligi qanday($N/m$)?
A) 25/8
B) 25/4
C) 8/25
D) 4/25

22. Rasmdan foydalanib ishqalanish koeffisientlari orasidagi munosabatni toping.




A) $\mu_1 > \mu_2 > \mu_3$
B) $\mu_2 > \mu_1 = \mu_3$
C) $\mu_1 = \mu_3 > \mu_2$
D) $\mu_3 > \mu_2 > \mu_1$

23. Rasmda ko‘rsatilgan $A$ nuqtadagi jismning og‘irligi qaysi yo‘nalishga mos tushadi.




A) 1
B) 2
C) 3
D) 4

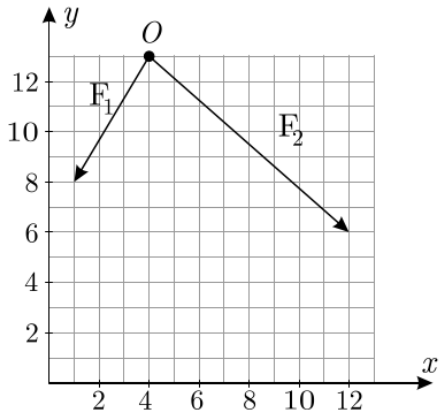
24. Agar yerning quyosh atrofida aylanish tezligi 30 $m/s$ ga teng bo‘lsa, noma’lum planetaning aylanish radiusi yernikidan 25 foizga kam bo‘lsa bu planetaning tezligi qanday?
A) 30,5
B) 34,6
C) 39
D) 41

25. Ishqalanish koeffitsienti 4 marta ortganda tormozlanish yo‘li qanday o‘zgaradi?
A) 2 marta ottadi
B) 4 marta kamayadi
C) 4 marta ortadi
D) 16 marta kamayadi

26. Massasi $m_1$ bo‘lgan jism $F$ kuch ta‘sirida 6 $m/s^2$ tezlanish bilan, shu kuch ta‘sirida $m_2$ massali jism 2 $m/s^2$ tezlanish bilan harakat qiladi. Bu kuch ta‘sirida $m_2 - m_1$ massali jism qanday tezlanish ($m/s^2$) bilan harakat qiladi?
A) 3
B) 2
C) 4
D) 1

27. To‘la sirti 96 $cm^2$ bo‘lgan kub shaklidagi jismning massasi 160 $g$ ga teng bo‘lsa uni zichligini ($kg/m^3$) toping.
A) 2,5
B) 2500
C) 3000
D) 1500

28. Rasmdan foydalangan holda jismga ta‘sir etuvchi natijaviy kuchni ($N$) toping. Har bir katak kvadrat va 1 $N$ ga teng.

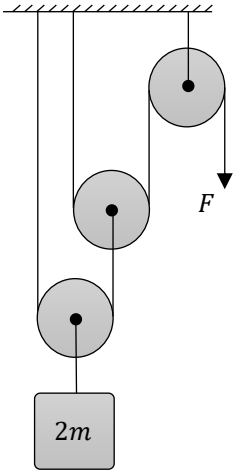



A) 12
B) 13
C) 6
D) 7

ISAK

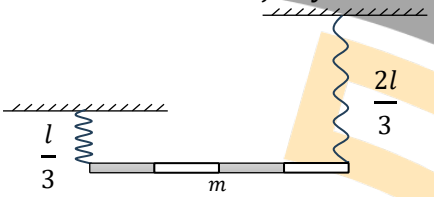
**STATIKA**

1. Rasmda berilgan bloklar sistemasidagi har bir blokning massasi $m$ ga teng bo‘lsa, $F$ kuchni toping.



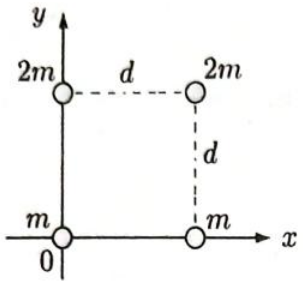

A) $\frac{mg}{4}$
B) $\frac{5mg}{4}$
C) $\frac{3mg}{2}$
D) $\frac{mg}{2}$

2. Vaznsiz prujinaga $m$ massali sterjen ilinganda u $x$ ga cho‘zildi. Agar bu prujinani $l/3$ va $2l/3$ bo‘lakka bo‘lib sterjen yana ilinsa, $2l/3$ uzunlikdagi prujina qanchaga cho‘ziladi?




A) $x/3$
B) $2x/3$
C) $3x/2$
D) $x$

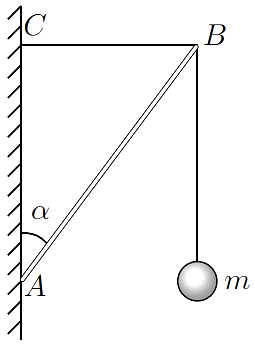
3. Tomonlari $d$ bo‘lgan kvadratning uchlariga rasmda ko‘rsatilgandek jismlar joylashtirilgan. Sistemaning massa markazi koordinatalarini toping.




A) $(\frac{d}{3}; \frac{d}{2})$
B) $(\frac{d}{2}; \frac{2d}{3})$
C) $(\frac{2d}{3}; \frac{d}{2})$

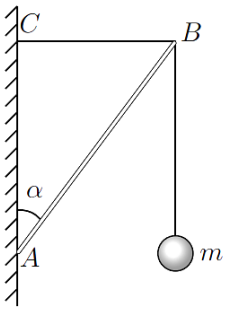
D) $(\frac{d}{3};\ d)$

4. Vaznsiz ipga massasi 12 $kg$ bo‘lgan sharcha rasmda ko‘rsatilgandek osilgan. Bunga ko‘ra $BC$ ipning taranglik kuchini $(N)$ toping. $sin\alpha = 5/13$.



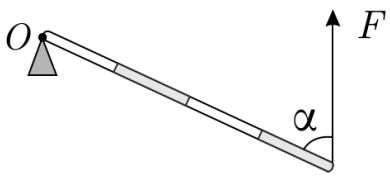

A) 50
B) 55
C) 60
D) 65

5. Agar $m = 12\ kg$ bo‘lsa, rasmdan foydalanib $AB$ sterjendagi reaksiya kuchini toping? $sin\alpha = 5/13$.




A) 165
B) 130
C) 155
D) 145

6. Chizmada massasi 2 $kg$ uzunligi 1,5 $m$ bo‘lgan sterjin berilgan. $F$ kuchni tayanchga nisbatan momentini $(N \cdot m)$ toping. $sin\alpha = 0{,}8$.




A) 6
B) 18
C) 9
D) 12

**SAQLANISH QONUNLARI**

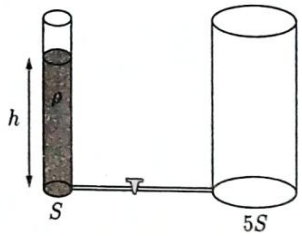
1. 7200 $m$ balandlikda 216 $km/h$ doimiy tezlik bilan uchayotgan samolyotning potensial energiyasi kinetik energiyasidan necha marta katta?

A) 30
B) 20
C) 10
D) 40

2. Impulsi 60 $kg \cdot m/s$ bo'lgan jism kinetik energiyasining 36 % ini yo'qotsa, impulsi qanchaga teng bo'lib qoladi ($kg \cdot m/s$)?

A) 48
B) 12
C) 36
D) 24

3. Asos yuzalari $S$ va $5S$ bo'lgan slindrik idishlar rasmda ko'rsatilgandek tublari birlashtirilgan. Agar kran ochilsa, qancha energiya issiqlikka aylanadi? $\rho$ — suyuqlik zichligi.




A) $\frac{1}{3}\rho gSh^2$
B) $\frac{5}{12}\rho gSh^2$
C) $\frac{2}{3}\rho gSh^2$
D) $\frac{5}{6}\rho gSh^2$

4. Kemaning uch dvigatel bor. Birinchi dvigatel kemaga 6 $m/s$ maksimal tezlik bera oladi. ikkinchisi 9 $m/s$. Uchala dvigatel bir vaqtda ishlasa kemaga maksimal 11 $m/s$ tezlik bera oladi. Uchinchi dvigatel kemaga qanday tezlik bera oladi? Suvning kema harakatiga qarshilik kuchi tezlikka proporsional.

A) 9
B) 8
C) 7
D) 2

5. Agar jism tezligi 3 $m/s$, kinetik energiyasi 6 $J$ ga teng bo'lsa, jisin impulsi ($kg \cdot m/s$) qanchaga teng?

A) 4
B) 16
C) 2
D) 18

6. Agar m massali jismning quvvati $N = \alpha t$ qonun bo'yicha chiziqli ortsa, jismning oniy tezligi qanday bo'ladi?

A) $\sqrt{\frac{\alpha}{m}}t$

B) $\sqrt{\frac{m}{\alpha}}t$

C) $\sqrt{\frac{2m}{\alpha}}t$

D) $\sqrt{\frac{2\alpha}{m}}t$

7. Massasi 0,5 $kg$ bo‘lgan sharcha 8 $m/s$ tezlik bilan janub tomonga, massasi 1 $kg$ bo‘lgan ikkinchi sharcha esa 3 $m/s$ tezlik bilan g‘arbga tomon harakatlanmoqda. Agar sharlar o‘zaro noelastik to‘qnashsa, ularning to‘qnashuvdan keyingi tezligini $(m/s)$ toping.

A) 5/3

B) 10/3

C) 2/3

D) 14/3

8. Massasi $m_1$ va $m_2$ bo‘lgan jismlarning tezliklari mos ravishda $\vec{\vartheta}_1$ va $\vec{\vartheta}_2$ bo‘lgan jismlar noelastik to‘qnashdi. To‘qnashuvdan so‘ng birinchi jismning kinetik energiyasini aniqlang.

A) $\frac{m_1}{2}(\frac{m_1\vec{\vartheta}_2+m_2\vec{\vartheta}_1}{m_1+m_2})^2$

B) $\frac{m_1}{2}(\frac{m_1\vec{\vartheta}_1+m_2\vec{\vartheta}_2}{m_1+m_2})^2$

C) $\frac{m_1}{2}(m_1\vec{\vartheta}_2+m_2\vec{\vartheta}_1)^2$

D) $\frac{m_1}{2}(m_1\vec{\vartheta}_1+m_2\vec{\vartheta}_2)^2$

9. Massasi 2 $kg$, tezligi 6 $m/s$ va massasi 1 $kg$, tezligi 5 $m/s$ bo‘lgan jismlar o‘zaro tik harakatlanmoqda ularning natijaviy impulsi qanday$(kg\cdot m/s)$?

A) 13

B) 12/13

C) 25

D) 169

10. Agar prujinali mayatnikning to‘la energiyasi 5 marta oshsa tebranish amplitudasi qanday o‘zgaradi.

A) 5 marta kamayadi

B) 5 marta ortadi

C) $\sqrt{5}$ marta ortadi

D) $\sqrt{5}$ marta kamayadi

11. Yer sirtidan 6400 $km$ balandlikda aylanma harakat qilayotgan 4 $kg$ massali jismni kinetik energiyasini $(MJ)$ toping.

A) 64 B) 128 C) 32 D) 16

**BOSIM. GIDRO VA AERODINAMIKA**

1. Bino bir nechta granit ustunlarga ega bo‘lib, har bir ustunning hajmi 9 $m^3$ ga, asosining yuzi 1,4 $m^2$ ga teng. Granitning zichligi 2800 $kg/m^3$. Bitta ustun o‘z og‘irligi ta’sirida poydevorga qancha bosim ($kPa$) berishini aniqlang.
A) 160
B) 200
C) 180
D) 220

2. Suvning sirtidan 5 $m$ balandda quvvatli so‘ruvchi nasos o‘rnatilgan. Nasos suv sirti bilan quvur orqali birlashtirilgan. Nasos ishlagan paytda suvning bu quvurdagi maksimal tezligini ($m/s$) aniqlang.
A) 8
B) 6
C) 10
D) 9

3. Massasi 3 $kg$, zichligi esa 60 $kg/m^3$ bo‘lgan qutqaruv chambaragining suvdagi ko‘tarish kuchini aniqlang ($N$).
A) 470
B) 320
C) 400
D) 280

4. 80 $m$ chuqurlikdagi kemada 3 $cm$ diametrli teshik bor. 20 minutda kemaga qanday hajmli suv ($m^3$) kiradi? $\pi = 3$.
A) 34,2
B) 32,4
C) 27
D) 25

5. Tomoni 8 $cm$ bo‘lgan kub shaklidagi muz bo‘lagi dastlab suvga, so‘ngra kerosinga botirildi. Muzning suyuqliklarga botgan qism hajmlarining farqi ($cm^3$) topilsin.
A) 57,4
B) 51,2
C) 60,15
D) 58,2

6. Agar jismning 4450 $kg$ hajmi 0,5 $m^3$ bo‘lsa, u qanday moddadan tayyorlanagan?
A) po‘lat
B) alyuminiy
C) mis
D) temir

7. Suv sirtidan 80 $m$ chuqurlikda suzib yurgan suv osti kemasi tirqishiga suv qanday tezlik ($m/s$) bilan kiradi?

A) 20
B) 25
C) 40
D) 50

8. Gidravlik pressda birinchi porshen $F = 40\ N$ kuch ta'sirida 4 $cm/s$ tezlik bilan pastlamoqda. Ikkinchi porshen 0,5 $mm/s$ tezlik bilan ko'tarilsa, unga qo'yilgan kuchni $(N)$ toping.
A) 800
B) 1600
C) 400
D) 3200

9. $10^7\ Pa$ bosiin normal atmosfera bosimidan necha marta katta?
A) 10
B) 100
C) 50
D) 5

10. Ko'l sirtidan 40 $m$ chuqurlikdagi umumiy bosimni $(kPa)$ aniqlang. $P_o = 10^5\ Pa$.
A) 500
B) 400
C) 40
D) 50

11. Agar shar ichiga geliy gazi o'rniga teng miqdorda neon gazi bilan almashtirilsa, sharning ko'tarish kuchi qanday o'zgaradi? Shar qobig'ining massasini hisobga olmang. Havoning molyar massasi 29 $g/mol$ ga teng.
A) 20/9
B) 29/20
C) 29/4
D) 25/9

12. Teshik idishga har sekundda 0,3 $l$ suv quyilmoqda. Suv sathi 5 $sm$ ga teng bo'lgach, suv sathi ko'tarilishdan to'xtadi. Teshikning yuzini $(sm^2)$ toping.
A) 4
B) 2
C) 3
D) 5

ISAK

1. Suv sirtidagi to‘lqinning 1– va 3–do‘ngliklari orasidagi masofa 20 $cm$, uning tarqalish tezligi 0,1 $m/s$ ga teng. To‘lqinning tebranish davrini $(s)$ toping.
A) 1,2
B) 0,2
C) 0,5
D) 1

2. Birinchi va beshinchi do‘ngliklar orasidagi masofa 20 $cm$ ga teng. Agar to‘lqinning tarqalish tezligi 40 $cm/s$ bo‘lsa, uning davrini toping $(s)$
A) 1,25
B) 22,5
C) 0,125
D) 0,25

3. 25 $cm$ amphtuda va 2 $Hz$ chastota bilan tebranayotgan 10 $kg$ massali jismga qanday maksimal kuch $(kN)$ ta’sir etadi?
A) 0,4
B) 4
C) 16
D) 32

4. Matematik mayatnik birinchi marta $g/4$ tezlanish bilan tekis tezlanuvchan ko‘tarilgandagi tebranish davri $T_1$ ga, pastga $g/5$ tezlanish bilan tekis tezlanuvchan tushgandagi tebranish davri $T_2$ ga teng bo‘lsa, $T_1/T_2$ nisbatni toping.
A) 5/4
B) 4/5
C) 1
D) 3/2

5. To‘lqin uzunligi $\lambda$, fazalar farqi esa $4\pi/5$ bo‘lgan to‘lqinning yo‘llar farqi qanday?
A) $0{,}4\lambda$
B) $2\lambda$
C) $0{,}5\lambda$
D) $\lambda$

6. Tebranma harakat qilayotgan jismning harakat tenglamasi $x = Acos\omega t$ ga teng bo‘lsa, uning tezlanish tenglamasini toping.
A) $a = -A\omega^2 cos\omega t$
B) $a = -A\omega^2 sin\omega t$
C) $a = -A\omega cos\omega t$
D) $a = -A\omega sin\omega t$

7. Ikkita matematik mayatniklarning tebranish davrlari mos ravishda $T_1$ va $T_2$ ga teng. Shu ikki mayatnik iplari o‘zaro ulanib boshqa mayatnik hosil qilindi. Shu mayatnikning tebranish davrini

toping.

A) $\sqrt{T_1^2 - T_2^2}$

B) $\sqrt{T_1^2 + T_2^2}$

C) $T_1 - T_2$

D) $T_1 - T_2$

8. Davri 10 $s$ bo‘lgan ko‘ndalang to‘lqinning tarqalish tezhgi 40 $cm/s$ ga teng bo‘lsa, 10 $m$ masofada nechta to‘lqin uzunligi joylashadi?

A) 4

B) 5

C) 2

D) 2,5

9. $x = Acos\omega t$ qonuniyat bilan garmonik tebranayotgan jismning tezligi ifodasini ko‘rsating.

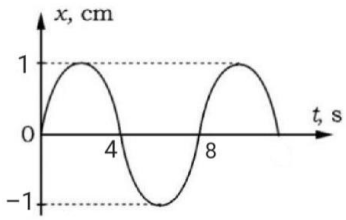
A) $-A\omega sin\omega t$

B) $-A\omega cos\omega t$

C) $-Asin\omega t$

D) $-Acos\omega t$

10. Tebranma harakat qilayotgan jism grafigidan foydalanib 3/8 sekundagi tezlanish topilsin.




A) $-\frac{\sqrt{3}\pi^2}{32}$

B) $\frac{\sqrt{3}\pi^2}{32}$

C) $\frac{\pi^2}{32}$

D) $-\frac{\pi^2}{32}$

11. $x = Asin\omega t$ tenglama bilan garmonik harakat qilayotgan jismning tezlik tenglamasini toping.

A) $\vartheta = A\omega cos\omega t$

B) $\vartheta = A\omega sin\omega t$

C) $\vartheta = -A\omega sin\omega t$

D) $\vartheta = Asin\omega t$

12. Tebranish chastotasi 200 $Hz$ bo‘lgan jismning tebranish davrini ($ms$) toping.

A) 5

B) 0,05

C) 5

D) 50

**MOLEKULYAR FIZIKA**

1. Ideal gazning bosimi $2{,}5 \cdot 10^5\ Pa$, zichligi 12 $kg/m^3$ ga teng. Gaz molekulalarining o‘rtacha kvadratik tezligini $(m/s)$ toping.
A) 160
B) 250
C) 300
D) 200

2. Yopiq idishdagi ozon $(O_3)$ 727 $°C$ haroratda turibdi. Qanchadir vaqtdan so‘ng barcha ozon kislorodga $(O_2)$ aylandi va idishdagi harorat 600 $°C$ ga pasaydi. Bunday o‘zgarishlardan so‘ng idishdagi bosim necha foiz kamayganligini aniqlang.
A) 60
B) 50
C) 30
D) 40

3. 19 $°C$ haroratli 5 $m^3$ havoda 34,5 $g$ suv bug‘i bor.

| $t, °C$ | $\rho, g/m^3$ |
|---|---|
| 15 | 12,8 |
| 16 | 13,6 |
| 17 | 14,5 |
| 18 | 15,4 |
| 19 | 16,3 |
| 20 | 17,3 |

Bu havo to‘yinishi uchun unga qancha $(g)$ massali suv bug‘latish kerakligini aniqlang.
A) 43
B) 47
C) 41
D) 45

4, Idishda teng miqdorda kislorod va geliy gazlari bor. Umurniy bosim 48 $kPa$ bo‘lsa, har bir gazning idish devoriga beradigan bosimini aniqlang $(kPa)$.
A) 12, 36
B) 16, 32
C) 4, 44
D) 24, 24

5. 15 $mol\ CH_4$ metanda nechta vodorod atomi bor?
A) $9 \cdot 10^{24}$ ta
B) $9 \cdot 10^{23}$ ta
C) $36 \cdot 10^{24}$ ta
D) $3{,}6 \cdot 10^{24}$ ta

6. Hajmi 60 $m^3$ va temperaturasi 17 °$C$ bo‘lgan xonada 400 $g$ suv bug‘latilsa, suv bug‘lari to‘yinishga erishadi. Xonadagi dastlabki nisbiy namlikni toping.

| $t, °C$ | $\rho, g/m^3$ |
|---|---|
| 15 | 12,8 |
| 16 | 13,6 |
| 17 | 14,5 |
| 18 | 15,4 |
| 19 | 16,3 |
| 20 | 17,3 |

A) 40 %
B) 64 %
C) 54 %
D) 48 %

7. O‘zgarmas massali ideal gazning hajmi $PV^3 = const$ qonuniyat bo‘yicha 4 marta oshdi. Gazning harorati qanday o‘zgargan?
A) 4 marta ortgan
B) 2 marta ortgan
C) 16 marta kamaygan
D) o‘zgarmagan

8. Bosimi 10 $MPa$ va harorati 770 $K$ bo‘lgan karbonat angidrid gazining zichligi ($g/m^3$) qanday?
A) 68,8
B) 57,9
C) 32
D) 62

9. $\rho_1$=2$\rho$ birinchi gaz $\rho_2$=6$\rho$ ikkinchi gaz bilan aralashtirildi va $\rho_{ar}$=4$\rho$ bo‘lgan aralashma hosil bo‘ldi. Quyidagi mulohazalarning qaysi biri to‘g‘ri?
1) birinchi gaz hajmi 2 marta katta
2) gaz hajmlari o‘zaro teng
3) birinchi gaz massasi ikkinchi gaz massasidan 2 marta katta
A) 1,3
B) faqat 2
C) 1,2
D) faqat 3

10. Teng miqdordagi ($\nu = const$) azot va kislorod gazlari ballonga qamalgan. Agar azot gazining bosimi $P$ ga teng bo‘lsa, ballondagi umumiy bosimni toping.
A) 3$P$
B) 2$P$
C) 5$P$

D) $P$

11. Normal sharoitda havo va kislorod gazlarining zichliklari $\rho_1$ va $\rho_2$ ga, havoning molyar massasi μ ga teng bo‘lsa, kislorodning molyar massasini toping.

A) $\frac{\rho_2 \mu}{\rho_1}$

B) $\frac{\rho_1 \mu}{\rho_2}$

C) $\frac{2\rho_2 \mu}{\rho_1}$

D) $\frac{\rho_2 \mu}{2\rho_1}$

12. Hajmi 60 $m^3$ bo‘lgan xonaning harorati 17 °$C$ va havoning nisbiy namligi 60 % ga teng. Xonadagi suv bug‘ining massasini ($g$) toping. 17 °$C$ haroratdagi to‘yingan suv bug‘ining zichligi 15,4 $g/cm^3$ ga teng.

A) 1108,8

B) 554,4

C) 277,2

D) 138,6

13. Sovun pufagining sirt yuzasi 4 marta ortsa, sirt energiyasi necha marta ortadi?

A) 4

B) 8

C) 2

D) 16

14. Absolyut deformatsiyasi 8 $mm$ bo‘lgan prujinaning nisbiy deformatsiyasi 0,2 % ga bo‘lsa, uning dastlabki uzunligini toping ($m$).

A) 8

B) 4

C) 2

D) 6

15. Mexanik kuchlanieh 200 $MPa$, yuzasi eaa 1,5 $mm^2$ bo‘Igan jismga ta’sir etuvchi kuchni ($N$) aniqlang.

A) 300

B) 30

C) 60

D) 3

16. Ko‘ndalang kesim yuzasi $S = 8\ mm^2$ bo‘lgan 10 $m$ uzunlikdagi sterjenning yung moduli 5 $MPa$ bo‘lsa, uning bikirligini ($N/m$) toping.

A) 4

B) 2

C)1

D) 5

17. Ballonda modda miqdorlari teng bo‘lgan azot, kislorod va vodorod gazlari bor. Agar azot gazining bosimi 14 $kPa$ bo‘lsa, ballondagi umumiy gaz bosimini ($kPa$) toping.
A) 28
B) 42
C) 84
D) 14

18. Normal sharoitda havo va kislorod gazlarining zichliklari $\rho_1$ va $\rho_2$ ga, havoning molyar massasi μ ga teng bo‘lsa, kislorodning molyar massasini toping.
A) $\frac{\rho_2 \mu}{\rho_1}$
B) $\frac{\rho_1 \mu}{\rho_2}$
C) $\frac{2\rho_2 \mu}{\rho_1}$
D) $\frac{\rho_2 \mu}{2\rho_1}$

19. 15 $mol\ CO_2$ ning massasi qancha ($g$) bo‘ladi?
A) 44
B) 0,66
C) 660
D) 220

20. Yopiq idishda 1,5 $MPa$ bosim ostida 27 °$C$ haroratli gaz bor. Gazning temperaturasi 60 °$C$ ga orttirildi. Bunda idishdan gazning 40 % i chiqib ketdi. Gazning keyingi bosimini ($MPa$) toping.
A) 0,92
B) 1,08
C) 0,96
D) 0,88

21. Hajmi 30 $m^3$ bo‘lgan xonadagi nisbiy namlik 60 % ga teng. Agar xona harorati 23 °$C$ ga teng bo‘lsa, xonadagi suv bug‘ining massasini ($g$) aniqlang.

| $t$, °c | $\rho_t$, $g/m^3$ |
|---|---|
| 23 | 20,6 |
| 24 | 21,8 |
| 25 | 23,0 |

A) 370,8
B) 392,4
C) 414
D) 207

22. Ko‘ndalang kesim yuzasi 8 $mm^2$ bo‘lgan 10 $m$ uzunlikdagi sterjenning Yung moduli 5 $MPa$ bo‘lsa, uning bikirligini ($N/m$) toping.
A) 4

B) 2
C) 1
D) 5

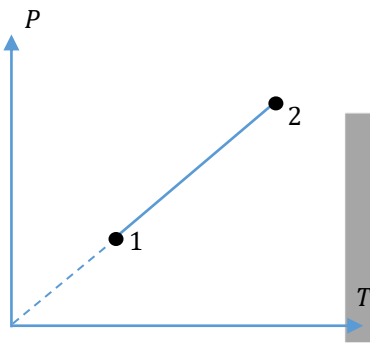
23. 16 $g$ misdagi atomlar sonini toping.
A) $1{,}5 \cdot 10^{23}$
B) $3 \cdot 10^{23}$
C) $1{,}5 \cdot 10^{22}$
D) $3 \cdot 10^{22}$

24. Ideal gaz bosimi $PV^2 = const$ qonuniyat bo'yicha ortmoqda. Bunda gazning temperatura qanday o'zgaradi?
A) ortadi
B) kamayadi
C) o'zgarmaydi
D) bosimga bog'liq emas

25. Isitkichning temperaturasi 627°C, sovutkichniki esa 600 $K$. Isitkich gazga 400 $J$ issiqlik miqdori beradi. Gazning uch siklda bajargan ishini $(J)$ toping.
A) 400
B) 300
C) 1200
D) 133

26. Suv ichidagi pufakcha hajmi 32 $mm^3$. Pufakcha ichidagi qo'shimcha bosimni $(Pa)$ aniqlang. $\pi$=3
A) 12
B) 24
C) 146
D) 73

27. Ideal gaz 1→2 o'tganda uning zichligi qanday o'zgargan?




A) izobarik ortgan
B) adiabatik kamaygan
C) izoxorik o'zgarmagan
D) izotermik ortgan

28. Yopiq idishda 1,5 MPa bosim ostida 27°C haroratli gaz bor. Gazning temperaturasi 60°C ga

orttirildi. Bunda idishdan gazning 40 % i chiqib ketdi. Gazning keyingi bosimini (MPa) toping

A) 0,92
B) 1,08
C) 0,96
D) 0,88

29. Hajmi 32 $mm^3$ bo‘lgan tomchi ichidagi qo‘shimcha bosim qanday Pa? π≈3.

A) 73
B) 35
C) 146
D) 57

30. 5 $mol\ CO_2$ gazidagi kislarod atomlari sonini toping.

A) $6 \cdot 10^{24}$
B) $3 \cdot 0^{24}$
C) $10^{24}$
D) $6 \cdot 10^{23}$

ISAK

1. 0,4 $kg$ massali alyuminiy berilgan. Uning issiqlik sig'imini $(J/K)$ aniqlang.
A) 900
B) 360
C) 640
D) 420

2. Asosining yuzi 1 $dm^2$ bo'lgan vertikal joylashgan silindrdagi 20 $kg$ massali porshen ostida geliy gazi bor. Geliy gazi izobarik qizdirilganda porshen 30 $cm$ ko'tarildi. Gaz kengayishda qancha ish $(J)$ bajarganini aniqlang. Atmosfera bosimi 100 $kPa$ ga teng.
A) 540
B) 360
C) 480
D) 240

3. Massalari 2 $kg$ va 3 $kg$ bo'lgan 83 $°C$ temperaturali mis bo'laklarini bir-biriga ishqalab eritib yuborish uchun qancha ish $(kJ)$ bajarish kerak? Misning ertish temperaturasi 1083 $°C$ ga teng. $c_{mis} = 380\ J/(kg \cdot K)$, $\lambda_{mis} = 205\ kJ/kg$.
A) 2485
B) 2925
C) 1025
D) 1900

4. Qandaydir jarayonda ideal gazning ichki energiyasi 16 $kJ$ ga ortdi. Keyin gaz adiabatik tarzda dastlabki haroratgacha sovitildi. Bunda tashqi kuchlarning bajargan ishini toping $(kJ)$.
A) $-10{,}67$
B) 16
C) $-16$
D) 10,67

5. Ballonda 2 $mol$ bo'lgan gaz qamalgan. Uning bosimi 1 $MPa$ va harorati 288 $K$. Gaz bosimini 4 marta oshirish uchun gazga qancha issiqlik $(kJ)$ berishi kerak?
A) 21,5
B) 25,7
C) 31
D) 43,2

6. $c = 2\ J/(g \cdot K)$ bo'lgan 4 $kg$ massali jismning issiqlik sig'imi qancha $(J/kg)$?
A) 8
B) 8000
C) 80000
D) 800

7. 5 $mol\ CH_4$ metan gazdan qanday energiya ajralishini hisoblang $(kJ)$. Bunda ajralgan issiqlikning 30 foizi tashqi muhitga tarqaladi deb oling.

A) 3456
B) 2464
C) 4634
D) 3698

8. Geliy gazining ichki energiya zichligi 18 $kJ/m^3$ ga teng. Uning bosimi ($kPa$) nimaga teng?
A) 24
B) 36
C) 18
D) 12

9. Gaz adiabatik kengayib 24 $kJ$ ish bajargan bo‘lsa, uning ichki energiyasi qanday o‘zgargan?
A) 24 $kJ$ ga kamaygan
B) 24 $kJ$ ga ortgan
C) 36 $kJ$ ga ortgan
D) o‘zgarmagan

10. Yopiq idishdagi argon gazining solishtirma issiqlik sig‘imini toping.
A) 265
B) 312
C) 345
D) 465

11. 5 $mol$ bir atomli gazning temperaturasi 200 $K$ ga ortsa, uning ichki energiyasi ($kJ$) qanday o‘zgaradi?
A) 8,31
B) 831
C) 12,465
D) 16,62

12. Yopiq idishdagi 6 $mol$ geliy gazining issiqlik sig‘imini ($J/K$) hisoblang.
A) 74,79
B) 78
C) 72,79
D) 60

13. 12 $mol$ bir atomli ideal gazning temperaturasi 60 $K$ ga ortdi. Uning ichki energiyasi qanday ($kJ$) o‘zgargan?
A) 3
B) 4
C) 9
D) 6

14. Xonadagi havo haroratini izobarik ravishda 290 K dan 300 K gacha isitildi. Ichki energiya qanday o‘zgargan?

A) chiziqli ortgan
B) chiziqli kamaygan.
C) harorat kvadratiga proporsional ortgan
D) o‘zgarmagan.

15. 4 mol ideal gazning ichki energiyasi 9600 J bo‘lsa absalyut harorat qanday K?
A) 193
B) 180
C) 210
D) 273

16. Mis bo‘lagi muz ustida turibdi mis muzga to‘la botishi uchun uning harorati qanday bo‘lishi kerak?
A) 72
B) 78
C) 88
D) 98

17. 20 $g$ geliy gazi temperaturasi o‘zgarmas bosim sharoitida $20°C$ ga ortrildi. Gazning bajargan ishini $(J)$ toping.
A) 831
B) 8,31
C) 83,1
D) 0,831

18. Ideal gazning hajmi $P/\sqrt{T} = const$ qonuniyat bo‘yicha 4 marta kamaydi. Bunda gazning ichki energiyasi qanday o‘zgaradi?
A) 2 marta ortadi
B) 16 marta kamayadi
C) 4 marta kamayadi
D) 8 marta oartadi

ISAK

**ELEKTROSTATIKA**

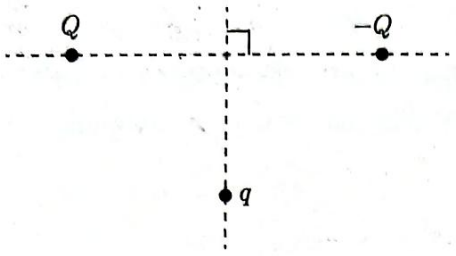
1. Zaryadli jismga ta'sir etuvchi 2 ta kuchning natijaviysi 32 $mN$ ga teng va gorizontal yo'nalgan. Agar og'irlik kuchi 24 $mN$ bo'lsa, elektr maydon ta'sirining modulini ($mN$) aniqlang.
A) 56
B) 12
C) 64
D) 40

2. Zaryadlari $3 \cdot 10^{-8}$ $C$ ga teng bo'lgan ikkita nuqtaviy zaryad havoda bir-biridan 50 $cm$ masofada turibdi. Ular 10 $cm$ gacha yaqinlashtirildi. Bu jarayonda bajarilgan ishni ($\mu J$) toping.
A) 32,4
B) 8
C) 4
D) 64,8

3. Sig'imi $C$ bo'lgan yassi havo kondensatoriga dielektrik singdiruvchanligi $\varepsilon$ bo'lgan modda kiritildi. Bu kondensatorda $C_x$ sig'imli kondensatorni ketma-ket ulaganda batareyaning umumiy sig'imi $C/2$ ga teng bo'ldi. $C_x$ sig'imni aniqlang.
A) $\frac{\varepsilon C}{2\varepsilon-1}$
B) $\frac{2\varepsilon C}{\varepsilon-2}$
C) $\frac{\varepsilon C}{\varepsilon-1}$
D) $\frac{2\varepsilon C}{\varepsilon-1}$

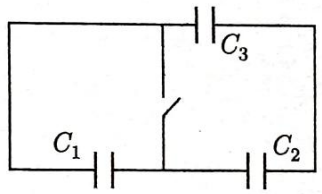
4. Zaryadlari $q$ dan bo'lgan $R$ va $2R$ radiusli sharlar bir-biriga tekkizilib dastlabki holatiga qaytarildi. Sharlarning o'zaro ta'sir kuchi qanday o'zgaradi?
A) 9/4 marta ortadi
B) 9/4 marta kamayadi
C) 9/8 marta kamayadi
D) 9/8 marta ortadi

5. Agar $q < 0$ va $Q > 0$ bo'lsa, q zaryadga ta'sir etuvchi kuch qanday yo'nalgan?




A) pastga
B) tepaga
C) chapga
D) o'ngga

6. Agar rasmdagi sxemada kalit ulansa quyidagi mulohazalarning qaysilari to'g'ri?




1) 1-kondensator maydon kuchlanganligi ortadi
2) 2-kondensator maydon kuchlanganligi kamayadi
3) 3-kondensator maydon kuchlanganligi o‘zgarmaydi

A) 1,2
B) faqat 2
C) faqat 3
D) 1,3

7. Misdan yasalgan sfera σ zichligi bilan zaryadlangan. Sfera radusi $R$ ga teng. Sfera sitidan $R/2$ masofadagi elektr maydon potensialini toping.

A) $\frac{2\sigma R}{3\varepsilon_0}$
B) $\frac{3\sigma R}{2\varepsilon_0}$
C) $\frac{2\sigma R}{\varepsilon_0}$
D) $\frac{4\sigma R}{3\varepsilon_0}$

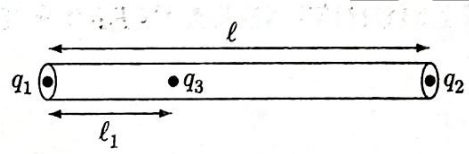
8. Zaryadlar orasidagi masofa 12 $cm$ ga orttirilganda, ular orasidagi ta’sir kuchi 9 marta kamaydi. Shu zaryadlar orasidagi dastlabki masofani $(cm)$ aniqlang.

A) 6
B) 12
C) 4
D) 16

9. Elektr maydon kuchlanganligi 40 $V/m$ bo‘lgan maydonda shu maydon tomonidan zaryadga 1600 $nN$ kuch ta’sir etayotgan bo‘lsa, shu zaryad $(nC)$ qiymatini toping.

A) 10
B) 20
C) 25
D) 40

10. $l$ uzunlikdagi quvurga bir xil ishorali zaryadlangan nuqtaviy zaryadlar rasmda ko‘rsatilgandek joylashtirilgan. Agar $q_3$ zaryad muvozanat holatida turgan bo‘lsa, $l_1$ masofani aniqlang.




A) $\frac{l\sqrt{q_1}}{\sqrt{q_1}+\sqrt{q_2}}$
B) $\frac{l\sqrt{q_2}}{\sqrt{q_1}+\sqrt{q_2}}$

C) $\frac{l\sqrt{q_1}}{\sqrt{q_1}-\sqrt{q_2}}$

D) $\frac{l\sqrt{q_2}}{\sqrt{q_1}-\sqrt{q_2}}$

11. Radiusi 5 $cm$ bo‘lgan metall sfera sim yordamida yerga ulangan. Metall sfera markazidan qanday masofada $-24$ $\mu C$ zaryad joylashtirilsa, sfera sirtida 8 $\mu C$ zaryad to‘planadi.

A) 20 cm
B) 15 cm
C) 10 cm
D) 5/3 cm

12. Dastlab tinch turgan $\alpha$ zarra tomon uzoqdan tezligi $\vartheta$ bo‘lgan proton markaziy yaqinlashmoqda. Ular orasidagi eng qisqa masofa nimaga teng?

A) $\frac{5ke^2}{m\vartheta^2}$
B) $\frac{4ke^2}{m\vartheta^2}$
C) $\frac{4ke^2}{m\vartheta}$
D) $\frac{5ke^2}{m\vartheta}$

13. Zaryadlari $-q, +2q, -8q$ va $+12q$ bo‘lgan to‘rtta bir xil simob tomchilari qo‘shildi. Hosil bo‘lgan tomchining umumiy zaryadini aniqlang.

A) $+12q$
B) $+6q$
C) $-3q$
D) $+5q$

14. Radiusi $5r$ va $3r$ bo‘lgan sharlarning zaryadlari mos ravishda $-20q$ va $+4q$ ga teng. Dastlab, sharlar tekkizilmasdan oldingi $F_1$ kuch bilan ta’sirlashmoqda. Ular bir-biriga tekkizilib qaytib olingandan keyin $F_2$ kuch bilan ta’sirlashadi. $F_1/F_2$ nisbatni aniqlang. Sharlar orasidagi masofa ularning radiuslaridan juda katta deb oling.

A) 4/3
B) 3/4
C) 12/5
D) 8/5

15. $Q$ va $q$ zaryadlar bir-biridan 10 sm masofada joylashgan. Zaryadlar orasidagi masofani 5 sm ga qisqartirish uchun 4/5 mJ ish bajarildi. Zaryadlar orasidagi masofani yana 3 sm ga qisqartirish uchun qanday ish bajarish kerak(mJ)?

A) 4,8
B) 2,4
C) 16/15
D) 8/15

16. O‘lchamlari bir xil 110 $V$ va $-40$ $V$ potensialga ega sharchalar bir biriga tekkazildi. Bunda birinchi

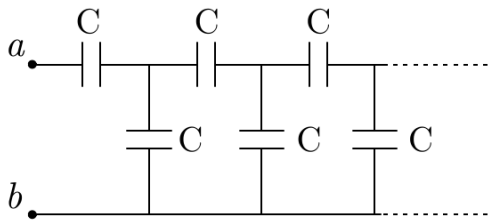
sharchaning potensiali qanday bo‘lib qoldi?
A) 70
B) 35
C) -5
D) 140

17. Agar jism 8 μC zaryadlangan bo‘lsa yo‘qotilgan elektronlar soni topilsin?
A) $5 \cdot 10^{12}$
B) $5 \cdot 10^{13}$
C) $5 \cdot 10^{11}$
D) $5 \cdot 10^{14}$

18. Zaryadi $q$ ga teng bo‘lgan sharning radiusi 4 marta ortsa uning energiyasi qanday o‘zgaradi?
A) 16 marta kamayadi
B) 4 marta kamayadi
C) 4 marta ortadi
D) 16 marta oartadi

19. Vakummda joylashgan nuqtaviy zaryadning o‘zidan $r$ masofada hosil qilgan maydon kuchlanganligi 8,82 $V/m$ ga teng bo‘lsa, uning kerosin ichida shu masofadagi maydon kuchlanganligi ($V/m$) qanchaga teng bo‘ladi?
A) 2,2
B) 1,1
C) 18,5
D) 4,2

20. Rasmdan foydalanib $a$ va $b$ nuqtalar orasidagi umumiy elektr sig‘imini toping.

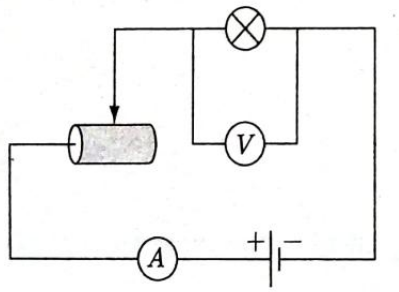



A) $\frac{\sqrt{5}+1}{2}C$
B) $\frac{\sqrt{2}+1}{2}C$
C) $\frac{\sqrt{5}-1}{2}C$
D) $\frac{\sqrt{3}-1}{2}C$

21. $C$ sig‘imli kondensator $U$ kuchlanishgacha zaryadlangan. Agar kondensator qoplamalari qarshligi $R$ bo‘lgan o‘tkzgich orqali ulansa, zaryadning 0,1 % qancha vaqtda o‘tadi?
A) $\frac{CR}{1000}$ B) $\frac{CR}{100}$ C) $\frac{R}{1000C}$ D) $\frac{C}{1000R}$

**O‘ZGARMAS TOK QONUNLARI. TURLI MUHITLARDA ELEKTR TOKI**

1. Reostatning surgichi chapga surilsa, ampermetrning ko‘rsatishi qanday o‘zgarishini aniqlang.




A) o‘zgarmaydi
B) kamayadi
C) ortadi
D) nolga teng bo‘ladi

2. Kimyoviy tok manbaining EYuKi 6 $V$, ichki qarshiligi 1/4 $\Omega$ ga teng. Manba zanjirga 2 $A$ tok berayotgan bo‘lsa, uning qutblaridagi kuchlanishni ($V$) toping.
A) 6,5
B) 7
C) 5,5
D) 5

3. Alyuminiydan yasalgan simning 0 °$C$ dagi qarshiligi 3,5 $\Omega$ ga teng. Uning qarshiligi 200 °$C$ da qanchaga ($\Omega$) ortishini aniqlang. $\alpha = 0{,}0028\ K^{-1}$.
A) 1,87
B) 1,96
C) 2,12
D) 3,14

4. Tok kuchi dastlab 10 $s$ da 0 dan $I$ gacha bir tekis ortadi, keyin 20 $s$ davomida o‘zgarmaydi, keyin 30 $s$ davomida bir tekis kamayadi. Shu bir daqiqa davomida simdan 400 $C$ zaryad o‘tgan bo‘lsa, $I$ ning miqdoti qanchaga teng($A$)?
A) 30
B) 20
C) 10
D) 5

5. Elektroliz jarayonida katodda 450 ta $Al^{+3}$ ioni ajralishi uchun, anodga nechta elektron yetib borishi kerak?
A) 450
B) 900
C) 1200
D) 1350

6. Anodga 2 $min$ davomida $24 \cdot 10^{16}$ ta elektron yetib borgan bo‘lsa, zanjirda hosil bo‘lgan tokni ($mA$) toping.
A) 0,16

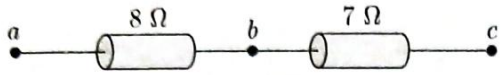
B) 0,32
C) 0,24
D) 0,12

7. Birinchi ampermetr qarshiligi 5 mΩ, ikkinchi ampermetr qarshiligi 15 mΩ. Bu ampermetrlar maksimal 9 $A$ tokni o‘lchashga mo‘ljallangan. Ularni parallel ulansa, qancha tok kuchini ($A$) o‘lchash mumkin?
A) $-18$
B) 12
C) 15
D) 27

8. $Cu^{+2}$ ionli eritma bor. Eritmadan $10^{23}$ ta elektron o‘tganda qanday massa ($g$) ajraladi?
A) 9,9
B) 5,3
C) 12
D) 4

9. EYuKi 6 $V$ bo‘lgan manbaga qarshiligi 20 Ω bo‘lgan rezistor ulangan. Agar rezistordagi kuchlanish 4 $V$ bo‘lsa, manbaning ichki qarshiligini toping (Ω).
A) 1
B) 5
C) 2,5
D) 10

10. $a$ va $c$ nuqtalar orasidagi potensiallar farqi 90 $V$ ga teng bo‘lsa, $b$ va $c$ nuqtalar orasidagi potensiallar farqini toping(V).




A) 21
B) 48
C) 24
D) 42

11. Qarshiliklari mos ravishda 5 mΩ va 15 mΩ bo‘lgan ampermetrlarning maksimal o‘lchash chegarasi bir xil 9 $A$ ga teng. Bu ampermetrlarni o‘zaro parallel ulab, qanday maksimal tok kuchini ($A$) o‘lchash mumkin?
A) 18
B) 12
C) 15
D) 9

12. Metall o‘tkazgichning $-20\ °C$ va $20\ °C$ haroratdagi qarshiliklari mos ravishda 11,2 Ω va 12 Ω ga teng. Shu o‘tkazgichning $300\ °C$ temperaturadagi qarshiligini (Ω) aniqlang.
A) 17,6

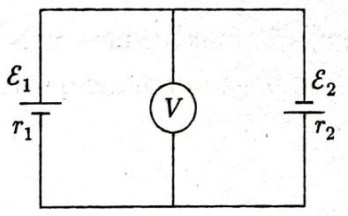
B) 21,6
C) 27,8
D) 15

13. Doimiy kuchlanish manbaiga ulangan o‘tkazgichdan 6 $A$ tok o‘tmoqda. Agar o‘tkazgichning 40 % i kesib tashlansa, undan qanday tok $(A)$ oqib o‘tadi?
A) 10
B) 36
C) 15
D) 20

14. Xonadonda muzlatkich 1,3 $kW$, elektr plita 1,4 $kW$, televizor 0,3 $kW$, umumiy quwati 1,4 $kW$ bo‘lgan 20 ta lampa 220 $V$ kuchlanish tarmog‘iga ulangan. Agar kechqurun 8 $A$ tok kuchida elektromobil zaryadlansa, xonadon simlari qanday tok kuchiga mo‘ljallangan bo‘lishi kerak?
A) 20
B) 28
C) 36
D) 42

15. Elektroliz vaqtida tok kuchi $I = 6 + 0{,}05t$ $[A]$ qonun bo‘yicha o‘zgarmoqda. 80 $s$ vaqt davomida qanday massali $(g)$ nikel ajralib chiqadi? $k = 0{,}35 \cdot 10^{-6}$ $kg/C$.
A) 0,224
B) 0,280
C) 0,488
D) 0,140

16. EYuK lari bir xil 3 $V$ dan bo‘lgan tok manbalari rasmda ko‘rsatilgandek ulangan. Agar tok manbalarining qarshiliklari 1 Ω va 3 Ω ga teng bo‘lsa, voltmetr ko‘rsatkichini aniqlang.




A) 4/3
B) 5/2
C) 3/2
D) 2/5

17. Ichki qarshiliklari 50 kΩ, va 80 kΩ bo‘lgan voltmetrlar maksimal 200 $V$ kuchlanishni o‘lchashga mo‘ljallangan. Ularni o‘zaro ketma-ket ulash orqali qanday maksimal kuchlanishni $(V)$ o‘lchash mumkin?
A) 400
B) 200
C) 325
D) 500

18. 2,5 $kW \cdot h$ elektr energiyasi necha $J$ ga teng?
A) $9 \cdot 10^6$
B) $9 \cdot 10^8$
C) $1{,}8 \cdot 10^6$
D) $7 \cdot 10^5$

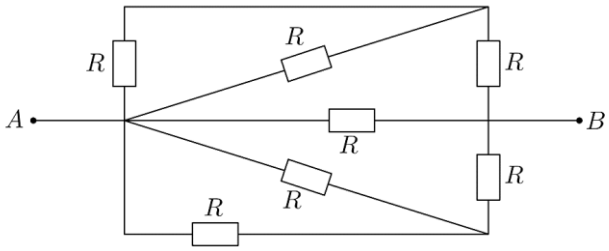
19. Elektr dvigateli orqali massasi 1 $t$ bo‘lgan yukni 360 $m$ masofaga ko‘tarish uchun necha so‘mlik elektr energiyasi sarflanadi? 1 $kW \cdot h$ elektr energiya narxi 300 so‘mga teng. Energiya isrofini hisobga olmang.
A) 450
B) 150
C) 350
D) 300

20. 120 $V$ kuchlanish manbaiga ulangan rezistordan 2,5 $min$ da 1,8 $kJ$ issiqlik miqdori ajralgan bo‘lsa, rezistordan oqib o‘tgan tok kuchini ($A$) aniqlang.
A) 0,1
B) 1
C) 5
D) 2

21. Har birining yuzasi 54 $cm^2$ dan bo‘lgan 30 ta qoshiq sirtini qalinligi 48 $\mu m$ bo‘lgan kumush qatlami bilan qoplash kerak. Agar qoshiqlardan 9,65 $A$ tok o‘tkazilgan bo‘lsa, ularni kumush bilan qoplash necha soat davom etgan? Kumushning valentligi 1 ga, molyar massasi 108 $g/mol$ ga va Faradey soni 96500 $C/mol$ ga teng.
A) 1,8
B) 7,2
C) 2,1
D) 3,6

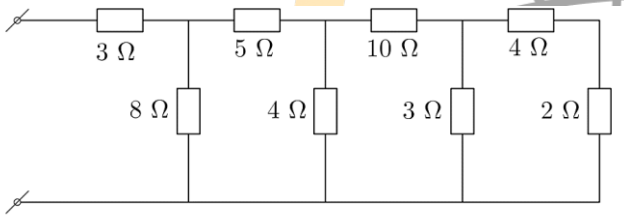
22. Alyuminiyning termik koeffitsiyentini aniqlang maqsadida doimiy kuchlanish manbaiga alyuminiy bo‘lagini ulab eriyotgan muzga solinganda zanjirdagi tok kuchi 35 $\mu A$ ga teng bo‘ldi. Qaynayotgan suvga solinganda esa tok kuchi 25 $\mu A$ bo‘ldi. Alyuminiyning termik koeffitsiyenti qanchaga ($10^{-3}\ K^{-1}$) teng?
A) 28,4
B) 36,4
C) 4
D) 12

23. $AB$ oraliqdagi umumiy qarshilikni toping.




A) $3R/10$
B) $2R/7$
C) $3R/7$
D) $3R$

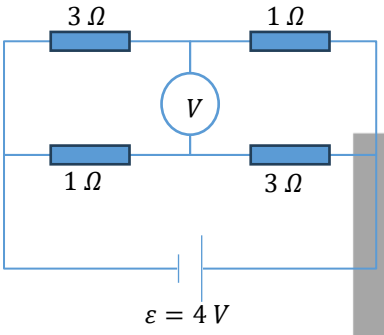
24. Umumiy qarshilikni toping.




A) 10
B) 2
C) 7
D) 3

25. 120 $V$ kuchlanish manbaiga ulangan rezistordan 2,5 $min$ da 3,6 $kJ$ issiqlik ajraldi, rezistordan oqib o‘tgan tok kuchi qanday?
A) 0,2
B) 2
C) 0,1
D) 4

26. Rasmdagi klemmalar orasidagi voltmeter necha voltni o‘lchaydi?




A) 2
B) 3
C) 2,5
D) 3,5

27. Alyuminiyning termik koefitsiyentini aniqlash maqsadida doimiy kuchlanishda dastlab eriyotgan

muzga solindi bunda tok kuchi 35 $mA$ teng bo‘ldi, keyin qaynayotgan suvga solindi bunda tok kuchi 25 $mA$ teng bo‘ldi, alyumininyning termik koefsentini toping ($10^{-3}$ $1/K$)

A) 28,4
B) 36,4
C) 4
D) 12

28. $Al^{+3}$ ionli elektroliz jarayonida $5 \cdot 96500$ $C$ ga teng zaryad oqib o‘tadi. Ajralib chiqqan moddaning miqdorini ($mol$) toping.

A) 5/3
B) 2/3
C) 3/5
D) 3/2

29. Anoddagi kuchlanish 48 $V$ ga teng. Anodga yetib borgan elektronni minimal kinetik energiyasi qanchaga ($eV$) teng.

A) 48
B) 96
C) 24
D) 12

ISAK

## ELEKTROMAGNETIZM

1. O‘lchamlari 2 $cm$ va 3 $cm$ bo‘lgan to‘g‘ri to‘rtburchak shaklidagi ramka induksiyasi 10 $T$ bo‘lgan magnit maydonda turibdi. Ramkadan 2 $A$ tok o‘tganda, unga ta’sir qiladigan maksimal kuch momentini ($mN \cdot m$) toping.
A) 96
B) 12
C) 48
D) 24

2. 8 Ω aktiv, 9 Ω sig‘im va 3 Ω induktiv qarshiliklarni ketma-ket holda ulab tuzilgan zanjir uchlariga effektiv qiymati 140 $V$ bo‘lgan o‘zgaruvchan kuchlanish berilgan. Zanjirdagi foydali quwatni ($W$) toping.
A) 1568
B) 1960
C) 1764
D) 588

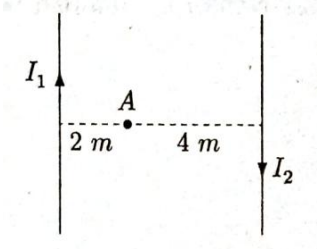
3. Elektron magnit maydonda $y$ o‘qi bo‘ylab harakatlanmoqda. Lorens kuchi $x$ o‘qiga qarshi yo‘nalgan bo‘lsa, magnit maydon yo‘nalishini aniqlang.
A) $x$ o‘qi bo‘ylab
B) $z$ o‘qi bo‘ylab
C) $y$ o‘qiga qarshi
D) $z$ o‘qiga qarshi

4. 200 $m$ to‘lqin uzunligida ishlayotgan radiostansiyani eshitish uchun 6 $MHz$ chastotaga sozlangan radiopriyomnik sxemasidagi sig‘imni qanday o‘zgartirish kerak?
A) 4 marta kamaytirish kerak
B) 16 marta oshirish kerak
C) 16 marta kamaytirish kerak
D) 4 marta oshirish kerak

5. $\frac{2\pi m\vartheta cos\alpha}{qB}$ kattalikning birligini toping.
A) $s$
B) $m$
C) $m/s$
D) $N$

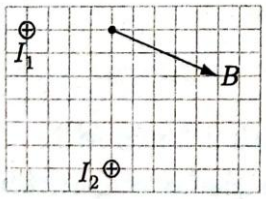
6. Rasmda qarama-qarshi oqayotgan to‘g‘ri toklar tasvirlangan. $I_1$ tokning $A$ nuqtada hosil qilgan induksiyasi 80 $T$ ga teng. Bunda $I_2 = 2I_1$ bo‘lsa, $A$ nuqtadagi natijaviy induksiyani ($T$) aniqlang.

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A) 0
B) 20
C) 100
D) 160

7. Berilgan chizmadan foydalanib $\frac{I_1+I_2}{I_2}$ nisbatni aniqlang. $B$ — natijaviy induksiya.

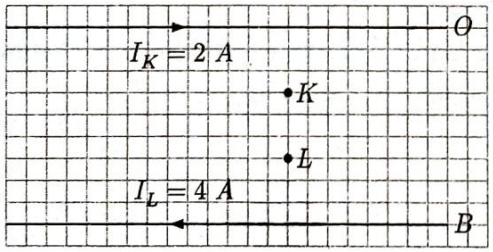



A) 14/3
B) 18/13
C) 19/15
D) 18/5

8. Uchta bir xil doimiy magnitning har birining magnit energiyasi $W$ ga teng. Agar bu magnitlarning mos qutblari jipslashtirib yopishtirilsa, bu sistemaning magnit maydon energiyasi qanday bo‘ladi?
A) $3W$
B) $6W$
C) $9W$
D) $W$

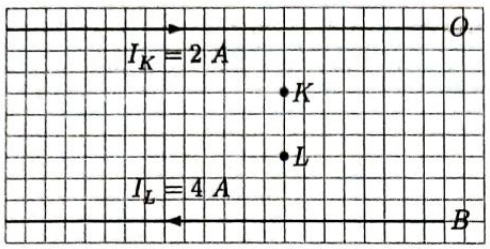
9. O‘zgaruvchan tok zanjiriga 8 Ω aktiv qarshilik, 2 Ω induktiv qarshilik va 8 Ω sig‘im qarshilik ketma-ket ulangan. Agar tok kuchining amplituda qiymati 0,6 $A$ ga teng bo‘lsa, zanjirdagi foydali quvvatni aniqlang ($W$).
A) 1,04
B) 0,84
C) 1,12
D) 1,44

10. Bir tekislikda joylashgan rasmdagi 0 va $B$ to‘g‘ri o‘tkazgichlardan mos ravishda 2 $A$ va 4 $A$ tok kuchi o‘tmoqda. O‘tkazgichlarning $K$ va $L$ nuqtalarda hosil qilgan natijaviy magnit induksiyalarining moduli $B_k$ va $B_l$ ga teng. Bunga asosan $B_k/B_l$ nisbat nimaga teng? Nuqtalar orasidagi masofalar o‘zaro teng.




A) 2/3
B) 3/4
C) 4/5
D) 5/4

11. Bir tekislikda joylashgan rasmdagi $X$ va $Y$ to‘g‘ri o‘tkazgichlardan mos ravishda 2 $A$ va 4 $A$ tok kuchi o‘tmoqda. O‘tkazgichlarning $K$ va $L$ nuqtalarda hosil qilgan natijaviy magnit induksiyalarning moduli $B_k$ va $B_l$ ga teng. Bunga asosan $B_l/B_k$ nisbat nimaga teng? Nuqtalar orasidagi masofalar o‘zaro teng.




A) 2/3
B) 3/4
C) 4/5
D) 5/4

12. $R$ radiusli halqadan $I$ tok oqmoqda. Agar halqa yarmi 90° qilib bukilsa, halqa markazidagi induksiya qanday o‘zgaradi?
A) $\sqrt{2}$ marta ortadi
B) $\sqrt{2}$ marta kamayadi
C) 2 marta kamayadi
D) 2 marta ortadi

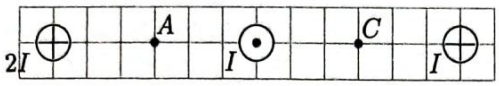
13. O‘zgaruvchan tok tarmog‘iga ketma-ket ravishda 6 Ω aktiv qarshilik, 3 Ω indiktiv va 11 Ω sig‘im qarshilik ulandi. Quvat koeffitsiyenti qanchaga teng?
A) 0,6
B) 0,8
C) 1
D) 0,96

14. Magnit oqimining birligi nima?
A) Genri
B) Tesla
C) Veber

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D) Joul

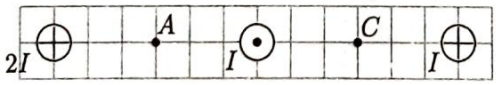
15. Rasmdan foydalangan holda $A$ nuqtadagi umumiy magnit maydon induksiya vektorini $C$ nuqtadagi umumiy magnit maydon induksiya vektoriga nisbatini $(B_A/B_C)$ toping.




A) 2
B) 1
C) 4
D) 3

16. Bir xil uzunlikdagi simdan kvadrat va halqa yasaldi va undan bir xil tok oqizildi. Agar ular bir jinsli magnit maydonga kiritilganda, halqaning maksimal aylantiruvchi momenti $M_1$ ga, kvadratniki esa $M_2$ ga teng bo'lsa, $M_1/M_2$ nisbatni aniqlang.
A) $4/\pi$
B) $\pi/2$
C) $\pi/4$
D) $\pi$

17. Rasmdan foydalangan holda $A$ nuqtadagi umumiy magnit maydon induksiya vektorini $C$ nuqtadagi umumiy magnit maydon induksiya vektoriga nisbatini $(B_A/B_C)$ toping.




A) 2
B) 1
C) 4
D) 3

18. Magnit maydon induksiya chiziqlariga tik holda uchib kirgan zarra massasi 2 marta ortsa, uning aylanma harakat radiusi qanday o'zgaradi?
A) 2 marta ortadi
B) 2 marta kamayadi
C) o'zgarmaydi
D) 4 marta ortadi

19. Tebranish konturi sig'imi 1,8 $\mu F$ bo'lgan kondensator va induktivligi 0,5 $mH$ bo'lgan g'altakdan iborat. Tebranish konturida hosil bo'layotgan elektromagnit to'lqinning tebranish davrini toping $(ms)$.
A) 0,19
B) 0,25
C) 1
D) 0,16

20. G'altakdan 1,6 $A$ tok kuchi oqib o'tganda unda 30 $Wb$ magnit oqimi vujudga keldi. Magnit oqimini o'zgartirmagan holda o'ramlar soni qanday o'zgartirilsa, 240 $mJ$ magnit maydon energiyasi hosil

bo‘ladi?
A) 100
B) 10
C) 20
D) 5

21. $4{,}5 \cdot 10^7$ $Hz$ chastotaga sozlangan qurilma 20 $m$ uzunlikdagi radio to‘lqinni qayd qilishi uchun sig‘imni necha marta oshirish kerak?
A) 3
B) 9
C) 27
D) 2,25

22. Bo‘shliqda elektromagnit to‘lqin tarqalmoqda. Agar elektr maydon kuchlanganligi 600 $kV/m$ ga teng bo‘lsa, elektromagnit maydon oqim zichligini ($W/m^2$) aniqlang.
A) 478
B) 956
C) 239
D) 119,5

23. Magnit maydonga ferromagnit o‘zak kiritilsa, magnit maydon qanday o‘zgaradi?
A) susayadi
B) biroz kuchayadi
C) o‘zgarmaydi
D) keskin kuchayadi

24. Radiuslari bir-biridan 3,5 marta farq qiluvchi halqalar o‘zaro perpendikulyar holda markazlari bir nuqtada qilib joylashtirilgan. Katta halqaning markazidagi magnit maydon induksiyasi 24 $mT$ va halqalar markazidagi natijaviy magnit maydon induksiyasi 25 $mT$ ga teng. Agar katta halqadan 36 $A$ tok oqayotgan bo‘lsa, kichik halqadan oqayotgan tokni ($A$) toping.
A) 10,5
B) 3
C) 4,3
D) 7,5

25. G‘altakdan 1,6 $A$ tok kuchi oqib o‘tganda unda 30 $Wb$ magnit oqimi vujudga keldi. Magnit oqimini o‘zgartirmagan holda, o‘ramlar soni qanday o‘zgartirilsa, 240 $mJ$ magnit maydon energiyasi hosil bo‘ladi?
A) 100
B) 10
C) 20
D) 5

26. Induktivligi 7,5 $H$ bo‘lgan g‘altak va sig‘imi 120 $pF$ bo‘lgan kondensator orqali tebranish konturi hosil qilindi. Konturda hosil bo‘lgan elektromagnit to‘lqinlarning to‘lqin uzunligini ($km$) aniqlang.

A) $12\pi$
B) $18\pi$
C) $6\pi$
D) $2\pi$

27. Tebranish konturida kondensatordagi zaryad $q = q_o sin\omega t$ qonuniyat bo‘yicha o‘zgarmoqda. G‘altak induktivligi $L$ bo‘lsa, konturdagi to‘la energiyani toping.
A) $\frac{1}{2}q_o L\omega$
B) $\frac{1}{2}q_o L\omega^2$
C) $\frac{1}{2}q_o L^2\omega^2$
D) $\frac{1}{2}q_o^2 L\omega^2$

28. Magnit maydonda aylanma harakat qilayotgan zaryadli zarrachaning aylanish chastotasi $\nu$ ga, Magnit maydon induksiyasi esa $B$ ga teng. Berilganlardan foydalanib zaryadning qanday kattaligini topsa bo‘ladi?
A) massasini
B) zaryad impulsini
C) solishtirma zaryadini
D) zaryadini

29. Induktiv qarshilik 5 Om aktiv qarshilik 4 Om, sig‘im qarshilik 2 Om bo‘lgan o‘zgaruvchan tok zanjiridagi effektiv kuchlanish 50 $V$ bo‘lsa foydali quvvatni hisoblang?
A) 400
B) 500
C) 420
D) 340

30. Ikkita zaryadli zarra magnit maydonda aylanma harakatlanib $2r$ va $3r$ radiusli aylana chizadi. Agar zarralarga ta'sir etuvchi Lorens kuchlari teng bo‘lsa, kinetik energiyalar nisbati $\frac{E_1}{E_2}$ topilsin.
A) 2/3
B) 4/9
C) 3/2
D) 9/27

31. Ikkita zaryadli zarra magnit maydonda aylanma harakatlanib $2r$ va $3r$ radiusli aylana chizadi. Agar zarralarning kinetik energiyalari teng bo‘lsa, Lorens kuchlari nisbati $\frac{F_1}{F_2}$ topilsin.
A) 3/2
B) 9/4
C) 2/3
D) 4

32. Tebranish konturida kondensator zaryadi kamaymoqda. Tok kuchining moduli qanday o‘zgaradi.
A) kamayadi

B) ortadi
C) o‘zgarmaydi
D) zaryadga bog‘liq emas

33. G‘altakdagi tok kuchi $I = 2{,}4 + 1{,}6t\ [A]$ qonuni bo‘yicha o‘zgarmoqda. Agar g‘altak induktivligi $L = 5\ mH$ bo‘lsa, unda hosil bo‘layotgan o‘zinduksion EYUK ni $(mV)$ toping.
A) 4
B) $-8$
C) 6
D) 2

34. Magnit maydon induksiyasi 0,25 $T$ bo‘lgan bir jinsli maydonga kiritilgan 20 $sm^2$ yuzaga ega bo‘lgan ramkadan 0,6 $A$ tok oqmoqda. Unga tasir etuvchi maksimal momentni $(N \cdot m)$ toping.
A) $3 \cdot 10^{-4}$
B) $4 \cdot 10^{-4}$
C) $10^{-4}$
D) $2 \cdot 10^{-4}$

ISAK

## OPTIKA. NISBIY NAZARIYASI.

1. To‘lqin uzunligi $\lambda$ bo‘lgan nur optik zichligi 4/3 bo‘lgan suyuqlikka tushib, sinmoqda. Bunda nurning chastotasi qanday o‘zgarishini aniqlang.
A) 25 % ortadi
B) 25 % kamayadi
C) o‘zgarmaydi
D) 33 % ortadi

2. Linza o‘zidan 40 $cm$ masofada joylashgan buyumning 2 marta kichiklashgan, mavhum tasvirini hosil qildi. Shu linzaning optik kuchini ($dptr$) aniqlang.
A) 2,5
B) -2,5
C) -7,5
D) 7,5

3. Fokus masofasi 18 $cm$ bo‘lgan linzada 4 marta kichiklashgan mavhum tasvir hosil qilindi. Buyum va tasvir orasidagi masofani aniqlang.
A) 67,5
B) 54,5
C) 40,5
D) 112,5

4. Lazer nuri ikkiga bo‘lindi. Ular bir xil qalinlikdagi suvda shishadan o‘tib ekranga tushadi. Ularning qaysi biri ekranga tezroq yetib boradi?
A) suvda
B) shishada
C) ikkalasida bir xil
D) muhitga bog‘liq emas

5. Fokus masofasa 18 $cm$ bo‘lgan yig‘uvchi linzada 4 marta kattalashgan to‘g‘ri tasvir hosil bo‘ldi. Buyumdan linzagacha bo‘lgan masofani ($cm$) aniqlang?
A) 13,5
B) 9
C) 22,5
D) 12

6. Difraksion panjara doimiysi $d = 1{,}4\ \mu m$. Panjara hosil qilgan difraksion manzaraning birinchi tartibli maksimumi 30° burchak ostida kuzatilayotgan bo‘lsa, nurning to‘lqin uzunligini ($cm$) toping.
A) 0,5
B) 0,8
C) 0,6
D) 0,7

7. Fokus masofasi 18 $cm$ bo‘lgan yig‘uvchi linzada 4 marta kattalashgan to‘g‘ri tasvir hosil bo‘ldi.

Linzadan buyumgacha bo‘lgan masofani ($cm$) aniqlang.
A) 27
B) 13
C) 35
D) 13,5

8. Optik kuchi $-6$ $dptr$ bo‘lgan ko‘zoynak taqadigan bolaning eng yaxshi ko‘rish masofasini toping($cm$).
A) 10
B) 5
C) 12
D) 15

9. Suyuqlik sirtiga yorug‘lik 36° burchak ostida tushmoqda. Agar sinish burchagi 24° bo‘lsa, singan va akslangan nurlar orasidagi burchakni toping.
A) 120°
B) 60°
C) 54°
D) 72°

10. Linzada buyumning 3 marta kichiklashgan, to‘g‘ri va mavhum tasviri hosil bo‘ldi. Agar buyum va tasvir orasidagi masofa 16 $cm$ bo‘lsa, linza fokusining modulini ($cm$) aniqlang.
A) 16
B) 12
C) 24
D) 4

11. Yassi ko‘zguga tushayotgan nurning yo‘nalishi 40° ga o‘zgarishi uchun, ko‘zguni qanday burchakka burish kerak?
A) 20
B) 10
C) 5
D) 40

12. Botiq ko‘zgudan 20 $cm$ masofada buyum joylashtirilgan. Agar ko‘zguda buyumning 5/3 marta kattalashgan tasviri hosil bo‘lgan bo‘lsa, ko‘zguning egrilik radiusini ($cm$) toping.
A) 50
B) 37,5
C) 25
D) 12,5

13. Tomoni 6 $cm$ bo‘lgan kvadrat ko‘zgu orqali qaralganda (ko‘z tekisligiga nisbatan) maksimal qanday yuzali sohani ($cm^2$) ko‘rish mumkin?
A) 36
B) 216

C) 144
D) 24

14. Difraksion panjara doimiysi 6 $\mu m$ ga teng. Ekranda 5 – tartibli maksimum nolinchi maksimumdan deyarli 90° ga farq qiladi. Panjaraga tushayotgan to‘lqinning uzunligini ($\mu m$) aniqlang.
A) 30
B) 1,2
C) 2,4
D) 4,8

15. Fokus masofasi 20 $cm$ bo‘lgan linzadan 60 $cm$ uzoqlikda mavhum va to‘g‘ri tasvir hosil qilindi. Linzaning kattalashtirishini aniqlang.
A) 4
B) 2
C) 7/15
D) 60/13

16. Yopiq zanjirdan tok o‘tishi natijasida magnit maydon hosil bo‘ldi. 5 $min$ da bu maydon qanday masofani bosib o‘tadi($km$)?
A) $9 \cdot 10^7$
B) $18 \cdot 10^7$
C) $15 \cdot 10^7$
D) $27 \cdot 10^7$

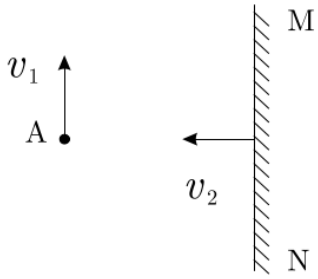
17. O‘quvchi 50° burchak ostida ko‘rinadigan jismga 10 marta kattalashtiradigan lupa orqali qaraganda burchak necha gradus bo‘lib ko‘rinadi?
A) 50°
B) 5°
C) 500°
D) 5000°

18. Botiq ko‘zgudan 20 $sm$ masofada buyum joylashtirildi. Agar buyumning 5/3 marta kattalashgan haqiqiy tasviri hosil bo‘lgan bo‘lsa. Ko‘zguning egrilik radiusi topilsin ($sm$).
A) 25
B) 12,5
C) 50
D) 20

19. Difraksion panjara doimiysi 6 $\mu m$. Bosh maksimumdan 5–tartibdagi maksimum 90° ostida ko‘rinadi. To‘lqin uzunlik qanday($\mu m$)?
A) 1,8
B) 1,2
C) 3
D) 2,4

20. Ko‘zguni qanday burchakka bursak qaytgan nur 40° ga buriladi?
A) 20
B) 80
C) 45
D) 30

21. Agar $A$ nuqta 8 $m/s$ tezlik bilan yassi ko‘zgu esa 3 $m/s$ tezlik bilan harakatlanayotgan bo‘lsa $A$ nuqta tasvirining yerga nisbatan tezligini toping.




A) 10
B) $\sqrt{52}$
C) 8
D) 13

22. Fokus masofasi 20 $sm$ bo‘lgan linzada buyumning to‘g‘ri va mavhum tasviri hosil qilindi. Linza va tasvir osrasidagi masofa 60 $sm$ ga teng. Linzaning kattalashtrishini aniqlang.
A) 2
B) 3
C) 4
D) 1/2

23. Fokus masofasi 20 $sm$ bo‘lgan linzada buyumning to‘g‘ri va mavhum tasviri hosil qilindi. Linza va tasvir osrasidagi masofa 60 $sm$ ga teng. Buyumdan linzagacha bo‘lgan masofani ($sm$) toping.
A) 15
B) 20
C) 25
D) 30

ISAK

## NISBIYLIK NAZARIYASI VA KVANT FIZIKASI

1. Ikki foton qarama-qarshi yo‘nalishda biri suvda ikkinchisi shishada harakatlanyapti. Ularning nisbiy tezligini toping.
A) $12c/17$
B) $17c/18$
C) $9c/17$
D) $7c/18$

2. Harakatdagi jismning massasi $m = (5/3)m_0$ ga teng. $m_0$ — jismning tinchlikdagi massasi. Jismning tezligini toping.
A) $0{,}4c$
B) $0{,}5c$
C) $0{,}6c$
D) $0{,}8c$

3. Fotoeffektning qizil chegarasi $\nu$ bo‘lgan metall sirtiga $5\nu$ chastotali, qizil chegarasi $2\nu$ bo‘lgan metall sirtiga esa $4\nu$ chastotali fotonlar tushmoqda. Uchib chiqqan fotoelektronlarning maksimal kiuetik energiyalari mos ravishda $E_1$ va $E_2$ bo‘lsa, $E_1/E_2$ ifodaning qiymatini toping.
A) $5/4$
B) $2$
C) $5/2$
D) $1/2$

4. Quyoshdan keladigan nurning intensivligi 1400 $W/m^2$ ga teng. Bu nurning absolyut qora sirtga bosimni qanday ($\mu Pa$)?
A) $17/4$
B) $14/3$
C) $14$
D) $7$

5. Berilgan jadvaldan foydalanib so‘roq o‘rnidagi maksimal kinetik energiyani ($eV$) toping.

| To‘lqin uzunlik | Maksimal kinetik energiya |
|---|---|
| $\lambda$ | 5 |
| $\lambda/2$ | 12 |
| $\lambda/3$ | ? |

A) 17
B) 19
C) 15
D) 21

6. Quvvati 150 $W$ bo‘lgan lampa nurlayotgan o‘rtacha to‘lqin uzunligi 2,2 $\mu m$. Lampadan $2{,}5\cdot10^{24}$ ta fotonlar necha sekundda ajraladi?
A) 1500

B) 3000
C) 4500
D) 2000

7. 8 $J$ necha $eV$ ga teng?
A) $5 \cdot 10^{19}$
B) $40 \cdot 10^{19}$
C) $8 \cdot 10^{19}$
D) 8

8. Metall sirtiga $0{,}75 \cdot 10^{15}$ $Hz$ chastotali fotonlar oqimi tushmoqda. Agar sirtga tushayotgan fotonlarning chastotasi 2 marta, tormozlovchi kuchlanish esa 3 marta ortgan bo‘lsa, metall uchun qizil chegarani $(nm)$ toping.
A) 400
B) 800
C) 200
D) 600

9. Yorug‘lik tezligiga yaqin tezlikda harakat qilayotgan zarraning to‘la energiyasi $E$ ga, impulsi $P$ ga teng bo‘lsa. Zarraning tinchlikdagi massasini toping.
A) $\frac{1}{c}\sqrt{\frac{E^2}{c^2} - P^2}$
B) $\frac{1}{c^2}\sqrt{\frac{E^2}{c^2} - P^2 c}$
C) $\frac{1}{c^2}\sqrt{\frac{E^2}{c^2} - P^2}$
D) $\sqrt{\frac{E^2}{c^2} - P^2}$

10. Tinchlikdagi niassasi $m$ bo‘lgan zarrachaning $0{,}8c$ tezlikdagi impulsini hisoblang.
A) $\frac{4mc}{3}$
B) $\frac{3mc}{4}$
C) $\frac{5mc}{4}$
D) $\frac{4mc}{5}$

11. Fotonning to‘lqin uzunligi 4 marta kamaysa, uning energiyasi qanday o‘zgaradi?
A) 2 marta ortadi
B) 2 marta kamayadi
C) 4 marta ortadi
D) 4 marta kamayadi

12. $\pi$ mezon ikkita yorug‘lik kvantiga ajraldi va o‘zaro qarama-qarshi yo‘nalishda harakatini davom ettirdi. Ularning energiyalari mos ravishda $\varepsilon_1$ va $\varepsilon_2$ ga teng ($\varepsilon_1 > \varepsilon_2$). $\pi$ mezonning ajralishdan oldingi impulsini aniqlang.

A) $\frac{\varepsilon_1+\varepsilon_2}{c}$
B) $\frac{\varepsilon_1-\varepsilon_2}{c}$
C) $\frac{\varepsilon_1-\varepsilon_2}{(\varepsilon_1+\varepsilon_2)c}$
D) $\frac{\varepsilon_1+\varepsilon_2}{(\varepsilon_1-\varepsilon_2)c}$

13. Tinchlikdagi zichligi 12,8 $g/cm^3$ bo‘lgan jism $0{,}6c$ tezlik bilan harakat qilmoqda. Uning harakatdagi zichligini aniqlang.
A) 20
B) 16
C) 9,6
D) 14,4

14. Elektron va pazitron juftligi annigliyatsiyalanganda 2 ta gamma kvanti hosil bo‘ldi. Dastlabki zarralarni boshlang‘ich tezliklari juda kichik bo‘lgan bo‘lsa, hosil bo‘lgan har bir gamma kvantining impulsini aniqlang. $m_e$ – elektron va pazitronning massasi.
A) $m_e c$
B) $2m_e c/h$
C) $m_e c/2h$
D) $m_e c/2$

15. Birinchi kosmik kema $2c/3$ tezlik bilan, ikkinchi kosmik kema unga qarama-qarshi ravishda $3c/4$ tezlik bilan harakatlanmoqda. Ularni nisbiy tezligini toping.
A) $17c/18$
B) $18c/17$
C) $17c/12$
D) $12c/17$

16. Muhitda ikki zarra bir biri tamon $0{,}7c$ va $0{,}4c$ tezlikda bir biri tamon harakat qilomoqda, ularni nisbiy tezligini toping.
A) $0{,}86c$
B) $0{,}95c$
C) $1{,}1c$
D) $0{,}3c$

ISAK

## ATOM VA YADRO FIZIKASI

1. $^{209}_{84}Po$ atomining tarkibini aniqlang.
A) 84 ta elektron, 209 ta neytron
B) 84 ta elektron, 125 ta neytron
C) 125 ta neytron, 84 ta proton
D) 209 ta nuklon, 84 ta neytron

2. Vodorod atomidagi elektron 4 — kvant holatidan 2 — kvant holatiga o‘tdi. Nurlangan fotonning energiyasini ($eV$) toping.
A) 2,55
B) 4,15
C) 6,28
D) 12,38

3. Beshinchi energetik sathgacha uyg‘otilgan elektron necha xil energiyali foton nurlashi mumkin?
A) 10
B) 5
C) 4
D) 8

4. Elektronning 4 — sathdan 2 — sathga o‘tgandagi va 4 — sathdan 3 — sathga o‘tgandagi chiqargan foton energiyalarining nisbatini aniqlang.
A) 27/7
B) 9/4
C) 3/2
D) 9

5. Atom yadrosida 11 ta proton va 13 ta neytron bor bo‘lsa, shu neytral atomning elektron qobig‘ida nechta elektron bor?
A) 11
B) 13
C) 24
D) 2

6. Atomdagi elektronning orbita radiusi 9 marta kamaysa, uning tezligi qanday o‘zgaradi?
A) 9 marta ortadi
B) 3 marta ortadi
C) 9 marta kamayadi
D) 3 marta kamayadi

7. Dastlab tinch turgan $\alpha$ zarra tomon uzoqdan tezligi $\vartheta$ bo‘lgan proton markaziy yaqinlashmoqda. Ular orasidagi eng qisqa masofa nimaga teng?
A) $\frac{5ke^2}{m\vartheta^2}$

B) $\frac{4ke^2}{m\vartheta^2}$

C) $\frac{4ke^2}{m\vartheta}$

D) $\frac{5ke^2}{m\vartheta}$

8. Vodorod atomi uchun birinchi bor orbita radiusi $0{,}53\cdot10^{-10}\ m$ bo‘lsa, ikkinchi va uchinchi orbita radiuslari farqini ($10^{-10}\ m$) toping.

A) 2,65

B) 2,25

C) 3,45

D) 6,89

9. Qandaydir element $\alpha$-yemirilishga uchrab $^{229}_{90}Th$ elementiga aylandi. Dastlabki elementdagi nuklonlar sonini aniqlang.

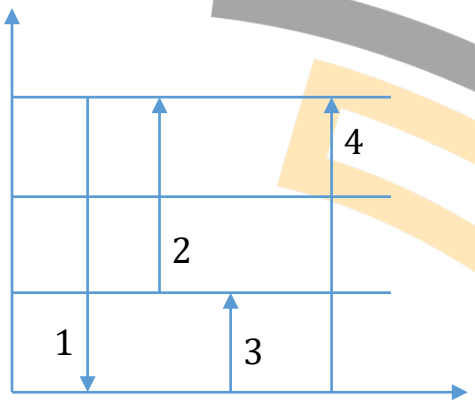
A) 233

B) 229

C) 225

D) 139

10. Qaysi birida eng katta impulsli foton yutadi?




A) 1

B) 3

C) 4

D) 2

11. $^{4}_{2}He$ geliy yadrosining bog‘lanish energiyasi 27 $MeV$ ga teng. Uni solishtrma bog‘lanish energiyasini ($MeV/nuklon$) toping.

A) 27/4

B) 27/2

C) 27

D) 54

12. Dastlabki atomlar soni $10^{20}$ ta. Moddaning yarim yemirilish davri 24 $h$. Agar har bir atom parchalanishida 400 $keV$ energiya chiqadi.30 sutkada qancha energiya chiqadi ($MJ$) ?

A) 5,3

B) 6,4
C) 4,8
D) 3,7

13. Agar $^{207}_{82}Pb$ neytral atomdagi elektonlar soni qanday?
A) 82
B) 207
C) 125
D) 94

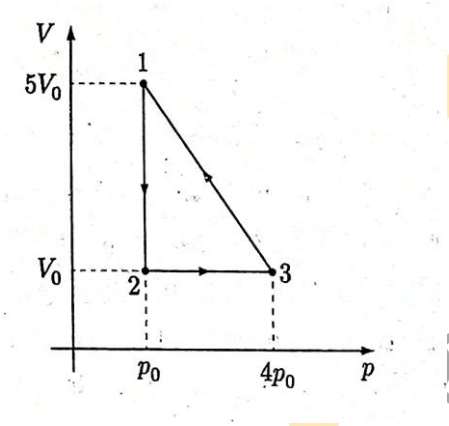
14. Vodorod atomida elektronning orbita radiusi 4 marta ortgan bo‘lsa, uning tezligi qanday o‘zgargan?
A) 4 marta ortgan
B) 2 marta ortgan
C) 4 marta kamaygan
D) 2 marta kamaygan

ISAK

## MOSLASHTIRISH TALAB QILINADIGAN SAVOLLAR

**1.**

**Topshiriqlar (33-35) va javob variant (A-F) larini o‘zaro to‘g‘ri moslashtiring.**




**33.** Gazning bajargan ishini toping.

**34.** 1 → 2 o‘tishda gaz qancha issiqlik chiqargan?

**35.** 3 → 1 o‘tishda gazga qancha issiqlik miqdori berilgan?

A) $11{,}5P_0V_0$

B) $8{,}5P_0V_0$

C) $12P_0V_0$

D) $6P_0V_0$

E) $10P_0V_0$

F) $2P_0V_0$

**2.**

**Topshiriqlar (33-35) va javob variant (A-F) larini o‘zaro to‘g‘ri moslashtiring.**

Idishda 2 kg suv va 0,5 kg muz termodinamik muvozanatda turibdi. Unga 424 °C temperaturali 1 kg alyuminiy solindi.

**33.** Muvozanat qaror topgandan so‘ng muz qancha issiqlik (kJ) oladi?

**34.** Muvozanat qaror topgandan so‘ng suv qancha issiqlik (kJ) oladi?

**35.** Muvozanat qaror topgandan so‘ng alyuminiy qancha issiqlik (kJ) oladi?

A) 364,5

B) 362

C) 346

D) 159,6

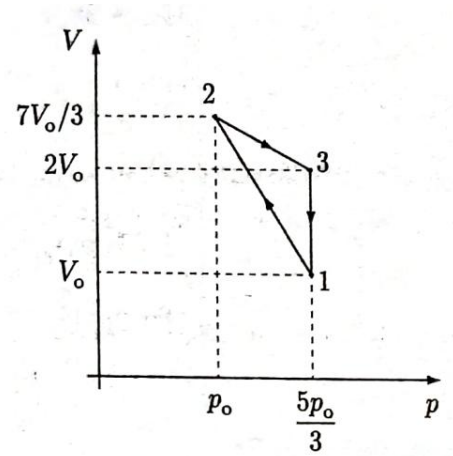
E) 204,9

F) 484

ISAK

3.

**Topshiriqlar (33-35) va javob variant (A-F) larini o‘zaro to‘g‘ri moslashtiring.**




**33.** Gazning bir siklda bajargan ishini toping.

**34.** $1 \to 2 \to 3$ o‘tishda gaz ichki energiyasi qanchaga o‘zgargan?

**35.** $1 \to 3 \to 2$ o‘tishda ichki energiya qanchaga ortgan?

A) $P_0V_0$

B) $\frac{5}{3}P_0V_0$

C) $\frac{1}{3}P_0V_0$

D) $\frac{7}{3}P_0V_0$

E) $\frac{1}{2}P_0V_0$

F) $\frac{5}{2}P_0V_0$

4.

**Topshiriqlar (33-35) va javob variant (A-F) larini o‘zaro to‘g‘ri moslashtiring.**

Raketa yuqoriga 2g tezlanish bilan 4 s harakatlanib, dvigatelini o‘chirdi.

**33.** Raketaning uchishni boshlaganidan 6 s dan keyingi tezligini (m/s) toping.

**34.** Raketaning dastlabki 6 s da bosib o‘tgan yo‘lini (m) toping.

**35.** Raketaning umumiy uchish vaqtini (s) toping.

A) 60

B) 80

C) 300

D) 16

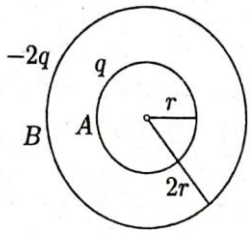
E) 22

F) 120

5.

**Topshiriqlar (33-35) va javob variant (A-F) larini o‘zaro to‘g‘ri moslashtiring.**

Misdan yasalgan radiuslari r va 2r bo‘lgan A va B sferalar o‘zaro konsentrik joylashtirilgan. Ularning zaryadlari mos ravishda q va -2q ga teng.




**33.** A sfera sirtidagi potensialni toping.

**34.** B sfera sirtidagi potensialni toping.

**35.** Agar A va B sferalar sim bilan tutashtirilsa, A sfera markazidan 3r masofadagi maydon potensialini aniqlang.

A) 0

B) $-k\frac{q}{3r}$

C) $-2k\frac{q}{r}$

D) $-k\frac{2q}{3r}$

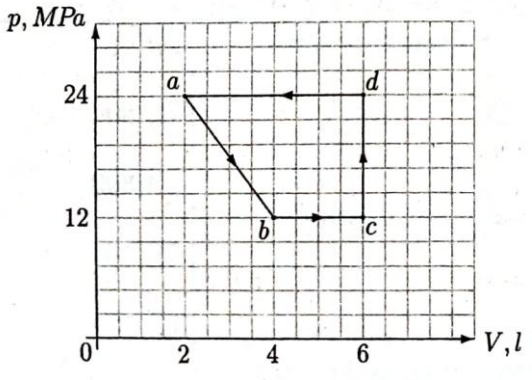
E) $-k\frac{q}{r}$

F) $-k\frac{q}{2r}$

6.

**Topshiriqlar (33-35) va javob variant (A-F) larini o‘zaro to‘g‘ri moslashtiring.**

Grafikda bir atomli ideal gaz bosimining hajmga bog‘lanishi keltirilgan. Bunda gaz ustida $a \to b \to c \to d \to a$ yopiq sikl bajariladi.




**33.** Gazning $a \to b$ o‘tishda bajargan ishini (kJ) toping.

**34.** $a \to b \to c$ o‘tishda gazga berilgan issiqlik miqdorini (kJ) toping.

**35.** $b \to c \to d$ o‘tishda gaz ichki energiyasining o‘zgarishini (kJ) toping.

A) 96

B) 120

C) 36

D) 44

E) 88

F) 144

7.

Topshiriqlar (33-35) va javob variant (A-F) larini o‘zaro to‘g‘ri moslashtiring.

Elektron va pozitron bir-biri bilan ta’sirlashib **pozitroniy** hosil qiladi. Bunda ular markazga nisbatan $r$ radiusli aylana bo‘ylab ta’sirlashadilar. $m$ – elektron massasi, $e$ – elektron zaryadi. (Gravitatsion ta’sirni inobatga olmang)

**33**. Pozitroniyning kinetik energiyasini toping.

**34**. Pozitroniyning to‘la energiyasini toping.

**35**. Pozitroniyning potensial energiyasini toping.

A) $\frac{ke^2}{4r}$

B) $-\frac{ke^2}{2r}$

C) $-\frac{ke^2}{4r}$

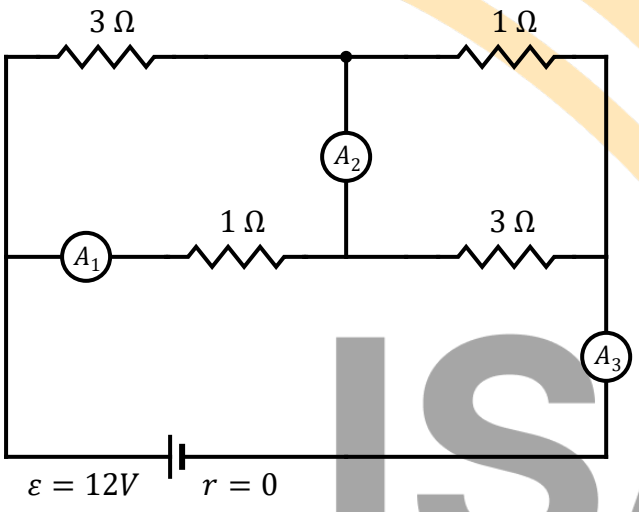
D) $mc^2-\frac{ke^2}{2r}$

E) $\frac{ke^2}{2r}$

F) $mc^2-\frac{ke^2}{4r}$

8.

Topshiriqlar (33-35) va javob variant (A-F) larini o‘zaro to‘g‘ri moslashtiring.
Quydagi berilgan sxemaga ko‘ra:




33. $A_1$ ampermetr ko‘rsatishi necha amper?

34. $A_2$ ampermetr ko‘rsatishi necha amper?

35. $A_3$ ampermetr ko‘rsatishi necha amper?

A) 2

B) 4

C) 1,5

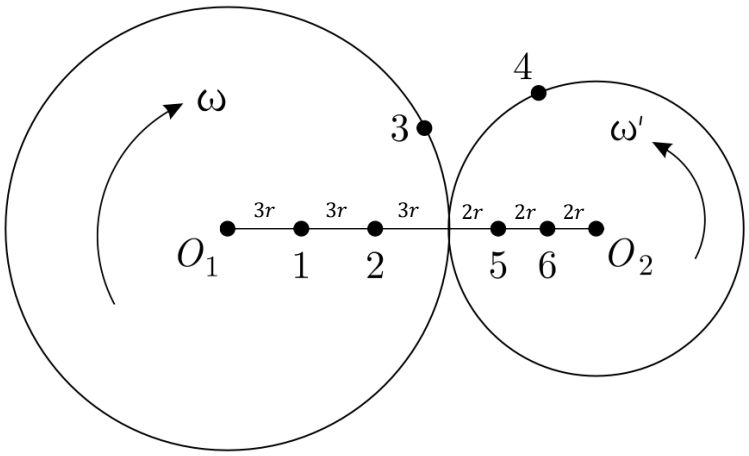
D) 6

E) 3

F) 8

9.

Topshiriqlar (33-35) va javob variant (A-F) larini o‘zaro to‘g‘ri moslashtiring.
Rasmda massalari teng bo‘lgan ikki diskning o‘zaro friksion bog‘lanishi tasvirlangan.




**33.** 1–6 nuqtalarning qaysi birida tezlanish eng katta?

**34.** 2, 4 va 6 nuqtalardan qaysi birining tezligi eng kichik?

**35.** 1–6 nuqtalarning qaysi biriga eng kichik kuch ta‘sir qiladi?

A) 1

B) 2

C) 5

D) 4

E) 6

F) 3

ISAK

## OCHIQ SAVOLLAR

1.
175 $m$ balandlikdan boshlang‘ich tezlik bilan erkin tashlangan jism 2–sekundda 25 $m$ yo‘l o‘tdi.
a) Jismning boshlang‘ich tezligini $(m/s)$ toping.
$Javob: a)$ _____
b) Jism dastlabki 4 $s$ da qancha $(m)$ masofa bosib o‘tishini aniqlang.
$Javob: b)$ _____

2.
150 $m$ balandlikdan boshlang‘ich tezlik bilan tashlangan jism ikkinchi sekundda 20 $m$ masofa bosib o‘tdi.
a) Jismning uchinchi sekund boshidagi tezligini toping $(m/s)$.
$Javob: a)$ _____
b) Oxirgi uch sekundda bosib o‘tgan masofasini toping $(m)$.
$Javob: b)$ _____

3.
Shamolning tezligi 12 $m/s$ bo‘lganda yomg‘ir tomchisi gorizont bilan 45° burchak hosil qiladi.
a) Shamol bo‘lmaganda tomchining tezligi $(m/s)$ qancha bo‘lgan?
$Javob: a)$ _____
b) Yomg‘ir tomchisi gorizont bilan 30° burchak hosil qilgan paytdagi shamolning tezligini $(m/s)$ toping.
$Javob: b)$ _____

4.
$r$ radiusli aylana bo‘ylab tinch holatdan harakatlana boshlagan jismning tezligi $\vartheta = kt$ qonuniyat bo‘yicha ortadi.
a) $2t$ vaqt davomida jismning bosib o‘tgan masofasini toping.
$Javob: a)$ _____
b) To‘la tezlanish ifodasini toping.
$Javob: b)$ _____

5.
Vaznsiz qo‘zg‘almas blok orqali o‘tkazilgan ingichka ipning uchlariga og‘irliklari $P$ va 20 $N$ bo‘lgan yuklar osilgan. Yuklar tinch holatdan boshlab harakatga kelganidan 2,5 $s$ vaqt o‘tgach 20 $N$ og‘irlikdagi yuk 6,25 $m$ masofaga pastga tushdi.
a) Yuklarning tezlanishini $(m/s^2)$ toping.
$Javob: a)$ _____
b) Harakat vaqtida blok o‘qiga ta’sir etuvchi kuchni $(N)$ aniqlang.
$Javob: b)$ _____

ISAK

6.

Tinch turgan jism gorizontal 400 $N$ kuch ta'sirida 8 $m$ masofaga ko'chirildi. So'ngra erkin harakatlanib 32 $m$ masofa bosib to'xtadi.

a) Harakat davomidagi ishqalanish kuchini toping ($N$).

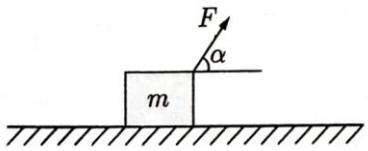
$Javob: a)$ _____

b) Tezlanuvchan harakatda ishqalanish kuchi bajargan ishni toping ($J$).

$Javob: b)$ _____

7.

Massasi $m = 2\ kg$ bo'lgan jism $F = 10\ N$ kuch ta'sirida tekis harakat qilmoqda. $sin\alpha = 0{,}6$.




a) Ishqalanish koeffitsiyentini toping.

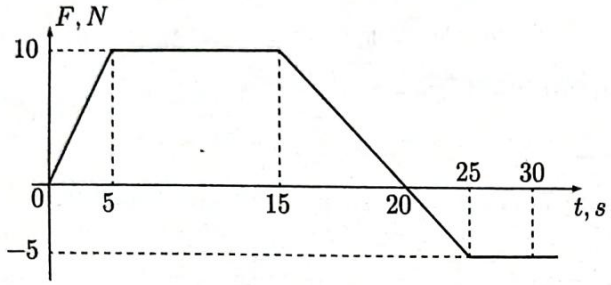
$Javob: a)$ _____

b) 10 m masofada ishqalanish kuchining bajargan ishini toping ($J$).

$Javob: b)$ _____

8.

Quyida massasi 5 $kg$ bo'lgan jismga ta'sir etayotgan kuchnmg vaqtga bog'lanish grafigi tasvirlangan. Uning boshlang'ich tezligi 2,5 $m/s$ ga teng.




a) Jismning 30 $s$ dan keyingi tezligini ($m/s$) toping.

$Javob: a)$ _____

b) 30 $s$ dan keyingi kinetik energiyasini ($J$) toping.

$Javob: b)$ _____

9.

Balandligi 12 $m$ ga teng bo'lgan qiya tekislik bo'ylab massasi 12 $kg$ ga teng bo'lgan jismni qiyalik yuqorisiga olib chiqish uchun 1800 $J$ ish bajarildi.

a) Ishqalanish kuchiga qarshi qancha ish ($J$) bajarilgan?

$Javob: a)$ _____

b) Jism qo'yib yuborilsa, qiya tekislik oxiridagi kinetik energiyasi qanchaga teng bo'ladi ($J$)?

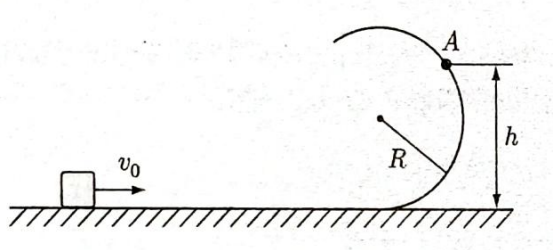
$Javob: b)$ _____

10.
Massasi 4 $kg$ bo‘lgan tinch turgan jism 20 $N$ kuch bilan gorizontal tortib ketilmoqda.
a) Qancha vaqtdan so‘ng kinetik energiyasi 200 $J$ ga teng bo‘ladi ($s$)?
$Javob$: $a$) _____
b) 2 s dan keyin jism impulsi ($kg \cdot m/s$) qanday bo‘ladi?
$Javob$: $b$) _____

11.
Jismga qiyaligi 45° bo‘lgan qiya tekislikka parallel tezlik berildi. Ishqalanish koeffitsiyenti 0,8 ga teng.
a) Jism qiya tekislik bo‘ylab ko‘tarilishdagi tezlanishi qanday ($m/s^2$)?
$Javob$: $a$) _____
b) Tushish vaqtining chiqish vaqtiga nisbati qanday?
$Javob$: $b$) _____

12.
Bikirligi 1000 $N/m$ bo‘lgan 75 ta prujina ketrna-ket ulandi va devorga mahkamlandi.
a) Prujinalar sistemasining umumiy biriklikni toping ($N/m$).
$Javob$: $a$) _____
b) Prujinalar sistemasining bir uchiga 80 $N$ kuch ta’sir etsa, umumiy uzayish qanday bo‘ladi ($m$) ?
$Javob$: $b$) _____

13.
Silliq yuzali tekislikdagi jismga biror $\vartheta_o$ boshlang‘ich tezlik berildi. U rasmda ko‘rsatilgan aylana bo‘ylab harakatlandi va yerdan $h$ balandlikdagi $A$ nuqtada tekislikdan uzildi. $R$ — aylana radiusi.




$a$) Jismning boshlang‘ich tezligini toping.
$Javob$: $a$) _____
$b$) $A$ nuqtadagi normal tezlanishni toping.
$Javob$: $b$) _____

14.
Prujinaga 40 $N$ cho‘zuvchi kuch ta’sir etganda uning uzunligi 30 $cm$ ga, 20 $N$ siquvchi kuch ta’sir etganda esa uning uzunligi 15 $cm$ ga teng bo‘ldi.
a) Prujinaning boshlang‘ich uzunligini ($cm$) toping.
$Javob$: $a$) _____
b) Prujinaning 2 $kg$ yuk osilgandagi uzunligini ($m$) toping.
$Javob$: $b$) _____

15.
Kosmik kema yer atrofida $r$ radiusli orbita bo‘ylab harakatlanmoqda. Yerning massasi $M$ ga teng.
a) Kosmik kema tezligini toping.
$Javob: a)$ _____
b) Kosmik kemaning aylanish davrini toping.
$Javob: b)$ _____

16.
Radiaktiv snaryad massalari $m_1$, $m_2$, $m_3$ va tezliklari mos ravishda $\vartheta_1$, $\vartheta_2$, $\vartheta_3$ bo‘lgan uchta bo‘lakka bo‘lindi.
a) Snaryadning bo‘linmasdan oldingi impulsini toping.
$Javob: a)$ _____
b) Snaryadning bo‘linishdan oldingi tezligini toping.
$Javob: b)$ _____

17.
Yer sirtida turgan matematik mayatnik ipining uzunligi 0,5 $m$ ga teng. Bu mayatnik massasi va radiusi Yernikidan 2 marta katta bo‘lgan $Y$ sayyoraga ko‘chirildi.
a) $Y$ sayyora sirti yaqinida erkin tushish tezlanishini $(N/kg)$ aniqlang.
$Javob: a)$ _____
b) $Y$ sayyorada mayatnik yuqoriga 5 $m/s^2$ tezlanish bilan harakatlanayotgandagi tebranish davrining Yer sirtida tekis ko‘tarilayotgandagi tebranish davriga nisbatini toping.
$Javob: b)$ _____

18.
Silliq gorizontal sirt ustida uzunligi $l$ va bikirligi $k$ bo‘lgan vaznsiz prujinaning ikki uchiga $m_1$ va $m_2$ massali yuklar mahkamlangan.
a) $m_1$ jismdan sistemaning og‘irlik markazigacha bo‘lgan masofani aniqlang.
$Javob: a)$ _____
b) Sistemaning kichik tebranishlar chastotasini toping.
$Javob: b)$ _____

19.
Yer sirtida turgan matematik mayatnik ipining uzunligi 0,5 $m$ ga teng. Bu mayatnik massasi va radiusi yernikidan ikki marta katta bo‘lgan sayyoraga ko‘chirildi.
a) Sayyor sirti yaqinidagi erkin tushish tezlanishini aniqlang $(m/s^2)$.
$Javob: a)$ _____
b) Sayyorada mayatnik yuqoriga 5 $m/s^2$ tezlanish bilan harakatlanayotgandagi tebranish davrining Yer sirtida tekis harakatlanayotgandagi tebranish davriga nisbatini toping.
$Javob: b)$ _____

20.
Kuzatish boshlangandan 0,5 $s$ o‘tib, matematik mayatnikning kinetik energiyasi 3 $mJ$ dan 1 $mJ$ gacha kamaydi, potensial energiyasi esa 1 $mJ$ dan 3 $mJ$ gacha ortdi.
a) Mayatnikning tebranish davrini toping ($s$).
$Javob: a)$ _____
b) Kuzatish boshlangandan $T/6$ vaqt o‘tgach, uning potensial energiyasi qancha ($mJ$) bo‘ladi?
$Javob: b)$ _____

21.
1 $cm^3$ hajmli kub shaklidagi idishga 0,6 $cm^3$ kerosin va 0,4 $cm^3$ hajmda suv quyildi.
a) Suyuqliklarning idish tubiga beradigan umumiy bosimini toping ($Pa$) toping?
$Javob: a)$ _____
b) Idish tubidagi bosim kuchini ($mN$) toping?
$Javob: b)$ _____

22.
Hajmi 400 $l$ bo‘lgan yopiq idishda 27 °$C$ haroratli to‘yinmagan havo bor. Idishga 2 $g$ massali suv bug‘latilsa, idishdagi suv bug‘lari to‘yingan holatga o‘tadi. Suv bug‘ining to‘yingan holatdagi parsial bosimi 3,6 $kPa$ ga teng.
a) Dastlabki havoning absolyut namligini ($g/m^3$) toping.
$Javob: a)$ _____
b) Dastlabki havoning nisbiy namligini (%) toping.
$Javob: b)$ _____

23.
Massasi 0,5 $kg$ bo‘lgan geliy gazining temperaturasi 27 °$C$ ga teng. Unga izoxorik ravishda 25 $kJ$ issiqlik berildi.
a) Uning oxirgi temperaturasini toping ($K$).
$Javob: a)$ _____
b) Uning bosimi necha marta o‘zgargan?
$Javob: b)$ _____

24.
Sirt taranglik koeffitsiyenti 0,05 $N/m$ ga teng bo‘lgan suyuqlikka diametri 1 $mm$ bo‘lgan nay tushirildi. Suyuqlik zichligi 800 $kg/m^3$ ga teng.
a) Suyuqlik nayda qanday balandlikka ko‘tariladi ($cm$)?
$Javob: a)$ _____
b) Bu ko‘tarihshda sirt taranglik kuchi qancha ish bajaradi ($mJ$)?
$Javob: b)$ _____

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25.
Toshkentdagi eski aloqa teleminorasining umumiy massasi 1600 $t$ ga teng. Teleminorani har birining yuzasi 2 $dm^2$ dan bo‘lgan 4 ta asos ko‘tarib turadi. Asos materiali uchun Yung moduli $1{,}9 \cdot 10^{11}$ $Pa$ ga teng.
a) Asoslardagi kuchlanishni ($Pa$) toping.
$Javob$: $a$) _____
b) Mexanik kuchlanish sabab asoslardagi nisbiy siqilishni (%) toping?
$Javob$: $b$) _____

26.
Argon gazi molekulalarining tezligi 100 $m/s$ dan 200 $m/s$ gacha oshirildi.
a) Gaz molekulalarining bosimi qanday o‘zgargan?
$Javob$: $a$) _____
b) Har bir gaz molekulasining kinetik energiyasi qanchaga o‘zgaradi ($J$).
$Javob$: $b$) _____

27.
Yopiq idishda 12 $mol$ neon gazi bor.
a) Uning issiqlik sig‘imini toping ($J/K$).
$Javob$: $a$) _____
b) Uning solishtirma issiqlik sig‘imini toping ($\frac{J}{kg \cdot K}$).
$Javob$: $b$) _____

28.
1000 $mol$ gaz sig‘adigan polietilen paketda 6 $mol$ geliy gazi bor.
a) Uning issiqlik sig‘imini toping ($J/K$).
$Javob$: $a$) _____
b) Uning solishtirma issiqlik sig‘imini toping($\frac{J}{kg \cdot K}$).
$Javob$: $b$) _____

29.
Bir atomli ideal gazga 6 $kJ$ issiqlik miqdori berildi. Bu jarayonda gazning hajmi va absolyut temperaturasi $V = k\sqrt{T}$ qonuniyat bo‘yicha bog‘langan. Bu yerda k — o‘zgarmas kattalik.
a) Bu jarayonda gazning bajargan ishini ($J$) toping.
$Javob$: $a$) _____
b) Bu jarayonda gazning ichki energiyasi qanchaga ($kJ$) ortishini toping.
$Javob$: $b$) _____

30.
Ideal issiqlik dvigateli isitgichining temperaturasi sovutgichnikidan 5 marta katta. Agar har siklda 600 $J$ ish bajarilsa,
a) Isitgichning beradigan issiqlik miqdorini toping ($J$).
$Javob$: $a$) _____

b) Faqat sovutgichning temperaturasi 2 marta orttirilsa, sovutgichga beriladigan issiqlik miqdori qanchaga teng bo'ladi $(J)$?

$Javob: b)$ _____

31.

Adiabatik jarayonda 3 mol bir atomli gazning ichki energiyasi 60 $J$ ga ortdi.

a) Gaz qancha ish $(J)$ bajargan?

$Javob: a)$ _____

b) Gazning temperaturasi qanday $(K)$ o'zgargan?

$Javob: b)$ _____

32.

42 gramm oltin bo'lagi 120 $m$ balandlikdan qattiq sirtga tushdi. Bunda barcha energiya oltinning ichki energiyasiga aylandi.

a) Uning ichki energiyasi qanchaga $(J)$ ortgan?

$Javob: a)$ _____

b) Bunda oltiiming temperaturasi qanchaga $(K)$ ortadi?

$Javob: b)$ _____

33.

2 $g$ suv tarkibidagi atomlarning barcha elektronlari yulib olindi va 10 $km$ masofaga uzoqlashtirildi.

a) Yulib olingan elektronlar sonini toping.

$Javob: a)$ _____

b) Elektronlar va yadrolar orasidagi tortishish kuchini toping $(N)$.

$Javob: b)$ _____

34.

1200 $kg$ massali elektromobil akkumulyatorining EYuKi 400 $V$ ga teng. Elektromobil tezligini 30 $m/s$ dan 5 $m/s$ gacha kamaytirdi.

a) Bu jarayonda elektromobil akkumulyatorida qancha energiya $(kJ)$ to'planishini aniqlang.

$Javob: a)$ _____

b) Bu jarayonda elektromobil akkumulyatorida qancha $(kC)$ zaryad to'planishini aniqlang.

$Javob: b)$ _____

35.

Radiusi 30 $cm$ bo'lgan sharga $\vartheta = 10^6\ m/s$ tezlik bilan uchib kelayotgan elektron kelib tushadi.

a) Sharga ko'pi bilan nechta elektron kelib tushadi?

$Javob: a)$ _____

b) Sharning maksimal potensiali qancha $(V)$?

$Javob: b)$ _____

36.

Sig'imlari bir xil $C$ dan bo'lgan kondensatorlar $U$ va $3U$ kuchlanishgacha zaryadlangan. Agar ular qutblari teskari holatda ulansa,

a) Natijaviy kuchlanishni toping.

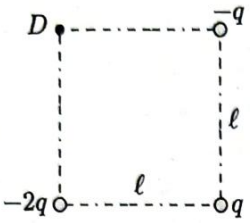
*Javob*: *a*) _____
b) Ajralib chiqqan issiqlik miqdorini toping.
*Javob*: *b*) _____

37.
1 $kW \cdot h$ energiya 300 *so'm*. Normal sharoitda 1 $m^3$ tabiiy gaz narxi 400 *so'm* ga teng.
a) 1 $MJ$ elektr energiyaning narxi necha so'm?
*Javob*: *a*) _____
b) 1 $MJ$ gaz narxi necha *so'm*?
*Javob*: *b*) _____

38.
Akkumulyatorning EYuKi 600 $V$ ga va zaryadi 450 $kC$ ga teng bo'lsa,
a) Tok kuchi 20 $A$ bo'lganda akkumulyator qancha vaqt ($h$) ishlaydi?
*Javob*: *a*) _____
b) Akkumulyatorda to'planadigan maksimal energiya necha $kW \cdot h$ ga teng?
*Javob*: *b*) _____

39.
Uchta nuqtaviy zaryad tomoni $a$ ga teng bo'lgan muntazam uchburchakning uchta uchida joylashtirilgan. Ularning zaryadlari mos ravishda $q_1$, $q$ va $q$ ga, teng.
a) $q_1$ zaryadga ta'sir etuvchi kuchni toping.
*Javob*: *a*) _____
b) $q_1$ zaryadning potensial energiyasini toping.
*Javob*: *b*) _____

40.
Tomonining uzunligi $l$ ga teng bo'lgan kvadratning uchta uchiga rasmda ko'rsatilgandek $-q$, $q$ va $-2q$ zaryadlar joylashtirilgan.




a) D nuqtadagi potensialni toping.
*Javob*: *a*) _____
b) Kvadrat markazidagi maydon kuchlanganligini toping.
*Javob*: *b*) _____

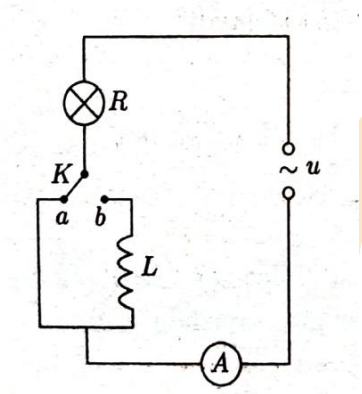
41.
1200 $kg$ massali elektromobil akkumulyatorining EYuKi 400 $V$ ga teng. Elektromobil tezligini 30 $m/s$ dan 5 $m/s$ gacha kamaytirdi.
a) Bu paytda elektromobil akkumulyatorida qancha energiya ($kJ$) to'planishini toping.
*Javob*: *a*) _____

b) Bu paytda elektromobil akkumulyatorida qancha zaryad $(C)$ to‘planadi?

*Javob*: *b*) _____

42.

Sxemada $R$=60 Ω, $L$=0,2 $H$ va $U = 160sin400t$ $[V]$.




a) Kalit $a$ vaziyatda bo‘lganida ampermetrning ko‘rsatishini $(A)$ toping.

*Javob*: *a*) _____

b) Kalit $b$ vaziyatda bo‘lganida ampermetrning ko‘rsatishini $(A)$ aniqlang.

*Javob*: *b*) _____

43.

Induktivligi 2 $H$ va sig‘imi 8 $pF$ bo‘lgan g‘altak va kondensatordan iborat tebranish konturi hosil qilindi.

a) Konturdagi tebranish chastotasini toping $(Hz)$

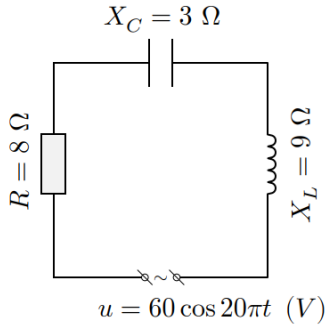
*Javob*: *a*) _____

b) Konturdagi elektromagnit to‘lqin uzunligini toping $(m)$.

*Javob*: *b*) _____

44.

O‘zgaruvchan elektr tarmog‘iga aktiv, sig‘im va induktiv qarshiliklar rasmda ko‘rsatilgandek ulangan.




a) Zanjirdagi tok kuchining effektiv qiymatini toping $(A)$?

*Javob*: *a*) _____

b) Elektr zanjirdan 5 $s$ da ajralgan issiqlik miqdorini toping $(J)$?

*Javob*: *b*) _____

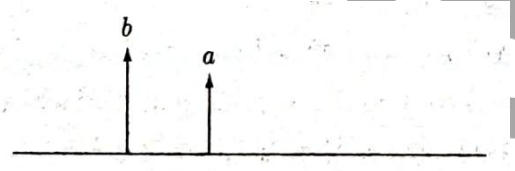
45.
Ikkita parallel tokli o‘tkazgichlar bir-birdan 5 $cm$ masofada joylashgan. Birinchi va ikkinchi o‘tkazgichdan bir xil yo‘nalishda oqayotgan toklar mos ravishda 4 $A$ va 3 $A$ ga teng.
a) Birinchi o‘tkazgichdan 4 $cm$, ikkinchi o‘tkazgichdan esa 3 $cm$ masofadagi natijaviy induksiyani toping ($\mu T$).
$Javob$: $a$) _____
b) Ularning 5 $m$ uzunlikdagi qismlari qanday kuch bilan ta’sirlashadi ($\mu N$)?
$Javob$: $b$) _____

46.
Induksiyasi $B = 10\ mT$ bo‘lgan bir jinsli magnit maydonga zarracha $\alpha = 60°$ burchak ostida $\vartheta_o$=1,5·$10^7$ $m/s$ tezlik bilan uchib kirdi.
a) Zarrachaning aylanish davrini toping ($ns$).
$Javob$: $a$) _____
b) Vint qadami uzunligini toping ($mm$).
$Javob$: $b$) _____

47.
Nuqtaviy manbadan elektromagnit to‘lqin tarqalmoqda. Undan 16 $m$ uzoqlikdagi nuqtada elektromagnit maydon energiya zichligi 2 $\mu J/m^3$ ga teng.
a) Manbadan 8 $m$ masofada energiya zichligi qanchaga teng ($\mu J/m^3$)?
$Javob$: $a$) _____
b) Manbadan 8 $m$ uzoqlikda intensivlik qanday ($W/m^2$)
$Javob$: $b$) _____

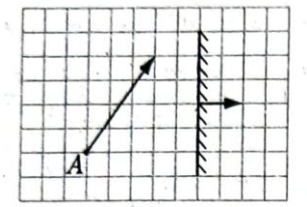
48.
Yorug‘lik nuri suvda $3d$, shishada $4d$ masofa bosib o‘tdi.
a) Yorug‘lik nurining suvda va shishada harakatlanish vaqtlari nisbatini toping.
$Javob$: $a$) _____
b) Yorug‘lik nurining butun harakatlanish vaqtidagi o‘rtacha tezligini toping.
$Javob$: $b$) _____

49.
Linza yordamida $a$ buyumning 5/4 marta kattalashgan $b$ tasviri hosil qilindi. Buyum va tasvir orasidagi masofa 2 $cm$ ga teng.




a) Linzaning optik kuchini ($dptr$) toping.
$Javob$: $a$) _____
b) Agar buyum linzaga 3 $cm$ yaqinlashtirilsa, linzaning kattalashtirishi nechaga teng bo‘lishini aniqlang.
$Javob$: $b$) _____

50.
Quyidagi rasmda $A$ nuqta va ko‘zguning harakat yo‘nalishi tasvirlangan. $A$ nuqtaning tezligi 5 $m/s$, ko‘zguning tezligi esa 2 $m/s$ ga teng.




a) Tasvirning yerga nisbatan tezligini toping($m/s$).
$Javob: a)$ _____
b) Tasvirning $A$ nuqtaga nisbatan tezligini toping ($m/s$).
$Javob: b)$ _____

51.
Intensivligi 24 $W/m^2$ bo‘lgan tabiiy yorug‘lik nuri qutblagichga tushmoqda. $sin\alpha = 0{,}8$.
a) Polyarizatordan o‘tgan nurning intensivligini toping ($W/m^2$).
$Javob: a)$ _____
b) Analizatordan o‘tgan nurning intensivligini toping ($W/m^2$).
$Javob: b)$ _____

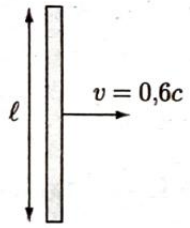
52.
Zarra $0{,}5c$ tezlik bilan harakatlanayotgandagi massasi $m/\sqrt{3}$ ga teng.
a) Zarraning tinchlikdagi massasini toping.
$Javob: a)$ _____
b) Zarraning $0{,}6c$ tezlik bilan harakatlanayotgandagi kinetik energiyasini toping.
$Javob: b)$ _____

53.
Jismning harakatdagi massasi tinchlikdagi massasidan 1,25 marta ortiq.
a) Uning tezligi qanday ($m/s$)?
$Javob: a)$ _____
b) Bunda uning zichligi necha marta ortadi?
$Javob: b)$ _____

54.
Tinchlikdagi massasi $m$ bo‘lgan jism $c/3$ tezlikda harakatlanmoqda. Jism tezligi ikki marta ortdi.
a) Jism impulsi qanchaga teng bo‘ladi ($kg \cdot m/s$)?
$Javob: a)$ _____
b) Impuls necha marta ortadi?
$Javob: b)$ _____

55.
Jismning tinchlikdagi massasi $10^{-7}\ kg$ va tezligi $0{,}8c$ ga teng.
a) Jismning klassik mexanika bo‘yicha impulsini hisoblang $(kg \cdot m/s)$
$Javob: a)$ _____
b) Uning relyativistiv impulsini hisoblang $(kg \cdot m/s)$.
$Javob: b)$ _____

56.
Rasmdagi jism o‘z uzunligiga tik ravishda harakatlanmoqda.




a) Uning $l$ uzunligi qanday o‘zgaradi?
$Javob: a)$ _____
b) Uning zichligi qanday o‘zgaradi?
$Javob: b)$ _____

57.
Kosmik kema $0{,}6c$ tezlik bilan harakat qilmoqda. Ikkinchi kosmik kema esa unga yetib olmoqda. Ularning nisbiy tezliklari $0{,}8c$ ga teng. $c$ — yorug‘likning vakuumdagi tezligi.
a) Ikkinchi kosmik kema tezligini klassik usulda toping.
$Javob: a)$ _____
b) Ikkmchi kosmik kema tezligini relyativistik (to‘g‘ri) usulda toping.
$Javob: b)$ _____

58.
$0{,}6c$ tezlik bilan harakatlanayotgan zarraning to‘la impulsi $P$ ga, to‘la energiyasi esa $E$ ga teng.
a) Zarraning $0{,}8c$ tezlik bilan harakatlangandagi to‘la energiyasini aniqlang?
$Javob: a)$ _____
b) Zarraning $0{,}8c$ tezlik bilan harakatlangandagi to‘la impulsini aniqlang?
$Javob: b)$ _____

59.
Tinchlikdagi massasi $10^{-8}\ kg$ bo‘lgan zarraning tezligi $0{,}6c$ ga teng.
a) Zarraning kinetik energiyasini $(J)$ klassik usulda toping?
$Javob: a)$ _____
b) Zarraning kinetik energiyasini $(J)$ to‘g‘ri (relyativistik) usulda toping?
$Javob: b)$ _____

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60.
Qaytarish koeffitsiyenti 0,5 bo‘lgan shaffof sirtga intensivligi 1200 $W/m^2$ bo‘lgan yorug‘lik tik tushmoqda.
a) Yorug‘likning sirtga beradigan bosimini toping ($\mu Pa$).
$Javob: a)$ _____
b) Shu yorug‘likning 40 $m^2$ yuzaga beradigan bosim kuchini toping ($\mu N$).
$Javob: b)$ _____

61.
Radioaktiv elementning yarim yemirilish doimiysi 0,014 $s^{-1}$ ga teng.
a) Yarim yemirilish davrini toping ($s$).
$Javob: a)$ _____
b) 2,5 minutdan keyingi yemirilgan atomlar ulushini toping (%).
$Javob: b)$ _____

62.
Birinchi radioaktiv preparatning aktivligi $A$ ga, yarim yemirilish davri 2 $h$ ga teng. Ikkinchi preparatning yadrolar soni uch marta ko‘pligiga qaramay aktivligi $A/2$ ga teng.
a) Ikkinchi preparatning yarim yemirilish davrini toping ($h$).
$Javob: a)$ _____
b) Ikkinchi preparatning 24 $h$ soatdan keyingi aktivligi nimaga teng bo‘ladi?
$Javob: b)$ _____

63.
Radioaktiv preparat ikkita radioaktiv moddadan tashkil topgan. Ularning yarim yemirilish davrlari $T$ va $2T$ ga, umumiy aktivlik esa $A$ ga teng. Qandaydir $t = 4T$ vaqtda umumiy aktivlik 12 marta kamaygan.
a) Radioaktiv moddalarning dastlabki atomlar soni nisbatini toping.
$Javob: a)$ _____
b) Keyingi atomlar soni nisbatini toping.
$Javob: b)$ _____

64.
$^{225}_{88}Ra$ yadrosi bitta pozitron va gamma yemirilishga uchradi.
$a)$ Hosil bo‘lgan yadroning zaryadi nechta elementar zaryadga ega.
$Javob: a)$ _____
$b)$ Hosil bo‘lgan yadrodagi neytronlar soni nechta?
$Javob: b)$ _____

65.
G‘alayonlangan vodorod atomining ionlashish energiyasi 3,4 $eV$ ga teng.
$a$) Atom foton yutib impuls momentini ħ ga oshirgan bo‘lsa, uning ionlashish energiyasi ($eV$) nechaga teng bo‘lishini aniqlang.
$Javob$: $a$) _____
$b$) Dastlabki atom foton nurlatib impuls momentini ħ ga kamaytirgan bo‘lsa, uning ionlashish energiyasi ($eV$) nechaga teng bo‘lishini aniqlang.
$Javob$: $b$) _____

66.
Shaffof idishda har xil energetik holatdagi vodorod atomlari bor. Uning ionlanish energiyasi 13,6 $eV$ ga teng.
$a$) Atom 1,89 $eV$ energiyali foton yutib ionlashmadi. Keyingi ionlanish energiyasini toping ($eV$).
$Javob$: $a$) _____
$b$) Atom 1,89 $eV$ energiyali foton nurlatsa, ionlanish energiyasi qanchaga teng ($eV$)?
$Javob$: $b$) _____

67.
Vodorod atomidagi elektron birinchi orbitadan boshqasiga o‘tganida uning kinetik energiyasi 16 marta kamaydi.
$a$) Elektronning aylanish radiusi qanday o‘zgargan?
$Javob$: $a$) _____
$b$) Elektronning aylanish davri qanday o‘zgargan?
$Javob$: $b$) _____

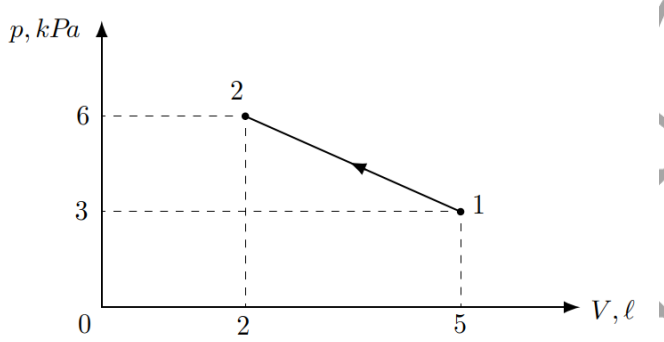
68.
Tennis raketkasiga koptok 12 $m/s$ tezlik bilan urillib 16 $m/s$ tezlikda qaytadi. Bunda koptokning kinetik energiyasi o‘zgarishi 14 $J$ ga teng bo’ldi.
$a$) Koptokning massasini toping($kg$).
$Javob$: $a$) _____
$b$) Koptok impulsining o‘zgarishini toping($kg \cdot m/s$).
$Javob$: $b$) _____

69.
Massasi 1,2 $kg$ bo‘lgam jism 80 $m$ balandlikdan 5 $m/s$ tezlik bilan gorizontal otildi.
$a$) Jismning uchish vaqtini toping($s$).
$Javob$: $a$) _____
$b$) Jismning 3 $s$ dan keyingi impuls o‘zgarishini toping($kg \cdot m/s$).
$Javob$: $b$) _____

70.
Gorizontga qiyalatib otilgan 800 $g$ massali jismning maksimal ko‘tarilish balandligi 80 $cm$ ga, uchish uzoqligi 2,4 $m$ ga teng.

$a)$ Jismning boshlang'ich tezligini toping$(m/s)$.
$Javob: a)$ _____
$b)$ Jismning maksimal balandlikdagi kinetik energiyasini aniqlang$(J)$.
$Javob: b)$ _____

71.
Bir atomli ideal gazning bosimi hajmga quyidagicha bog'langan.




$a)$ $1 \to 2$ o'tishda gaz ichki energiyasi qanchaga $(kJ)$ kamaygan?
$Javob: a)$ _____
$b)$ $1 \to 2$ o'tishda gaz atrofga qancha issiqlik $(kJ)$ ajratgan?
$Javob: b)$ _____

72.
Karno sikli bo'yicha ishlovchi issiqlik dvigatelida $\frac{T_1}{T_2} = \frac{7}{4}$ ga teng. Sovutgichga bir siklda 2400 $J$ issiqlik miqdori uzatiladi.
$a)$ FIK ni toping(%).
$Javob: a)$ _____
$b)$ 5 siklda bajarilgan ishni toping$(kJ)$.
$Javob: b)$ _____

73.
Ikki parallel cheksiz o'tkazgich orasidagi masofa 4 $cm$, ulardan oqadigan toklar esa mos ravishda 4 $A$ va 3 $A$ ga teng.
$a)$ Ikkala o'tkazgichning 5 $m$ uzunligi qanday kuch bilan $(\mu N)$ ta'sirlashadi?
$Javob: a)$ _____
$b)$ Birinchisidan 1 $cm$, ikkinchisidan esa 5 $cm$ uzoqlikda joylashgan nuqtadagi natijaviy induksiyani $(\mu T)$ aniqlang.
$Javob: b)$ _____

74.
Radiolokatordan chiqayotgan elektromagnit to'lqin uzunligi 12 $cm$ ga teng. U har sekundda 5000 ta impuls chiqaradi. Impulslar davomiyligi 2 $\mu s$.
$a)$ Radiolokatordan chiqayotgan elektromagnit to'lqinning tebranish davrini $(ns)$ toping.
$Javob: a)$ _____
$b)$ Elektromagnit to'lqin yetib borishi mumkin bo'lgan eng uzoq masofani $(km)$ toping.
$Javob: b)$ _____

75.
Tinchlikdagi energiyasi $E$ bo'lgan zarra tezligini $0{,}6c$ dan $0{,}8c$ gacha o'zgardi.
$a)$ Zarra impulsi o'zgarishini toping.

$Javob: a)$ _____
$b)$ Zarraning kinetik energiyasi qanchaga ortgan.
$Javob: b)$ _____

76.
Ikki kosmik kema bir tomonga mos ravishda $0{,}8c$ va $0{,}6c$ tezlikda harakatlanmoqda.
$a)$ Ularning nisbiy tezligini klassik usulda toping$(m/s)$.
$Javob: a)$ _____
$b)$ Ularning nisbiy tezligini relyativistik (to‘g‘ri) usulda toping$(m/s)$.
$Javob: b)$ _____

77.
Vodorod atomi elektroni $4-$ energetik sathdan $3-$ energetik sathga o‘tdi.
$a)$ Nurlangan foton chastotasini $(THz)$ toping.
$Javob: a)$ _____
$b)$ Nurlangan foton impulsini $(kg \cdot m/s)$ toping.
$Javob: b)$ _____

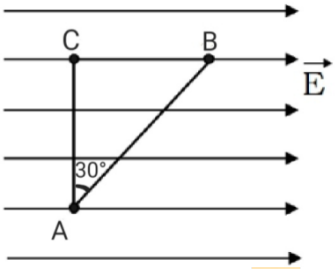
78.
Tomoni 2 $sm$ bo‘lgan ikkita kub shaklidagi muz bo‘lagi asosining yuzi 50 $sm^2$ va 45,6 $g$ suv bo‘lgan silindrik idishga tashlandi.
$a)$ Idish tubidagi gidrostatik bosim topilsin$(Pa)$.
$Javob: a)$ _____
$b)$ Idish tubidagi bosim kuchi topilsin$(N)$.
$Javob: b)$ _____

79.
Silindrik idishga teng massali kerosin va suv quyildi. Bunda suyuqliklar umumiy balandligi 27 $sm$ teng bo‘ldi.
$a)$ Suv ustunining balandligi qanday $(sm)$?
$Javob: a)$ _____
$b)$ Idish tubidan 8 sm balandlikdagi umumiy bosim qanday $(Pa)$?
$Javob: b)$ _____

80.
Ideal gazga izobarik 60 $kJ$ issiqlik berildi bunda uning hajmi 3 marta ortdi.
$a)$ Harorat o‘zgarishi qanday o‘zgaradi?
$Javob: a)$ _____
$b)$ Bajarilgan ishi qanday $(kJ)$?
$Javob: b)$ _____

81.
Issiqlik sig‘imi 500 $J/K$ bo‘lgan idishda 5 litr $10°C$ suv bor. Idishga 1 $kg$ massali $-5°C$ haroratli muz solindi.
$a)$ Natijaviy harorat qanday$(°C)$?
$Javob: a)$ _____
$b)$ Muz olgan issiqlik miqdori qanday$(kJ)$?
$Javob: b)$ _____

82.
Rasmda elektr maydon kuchlanganligi $E$ bo‘lgan bir jinsli maydonda proton $A$ nuqtadan $B$ nuqtaga ko‘chirildi. $AB = l$.




$a$) Bunda protonning kinetik energiyasi qanday o‘zgargan?
$Javob$: $a$) _____
$b$) $A - B - C - A$ o‘tishda maydon tomonidan qanday ish bajarilgan?
$Javob$: $b$) _____

83.
Uzunligi 12 $m$, qiyaligi 0,6 bo‘lgan qiya tekislikning yuqori nuqtasidan 5 $kg$ massali jism tusha boshladi.
$a$) Jismning qiya tekislik etagidagi tezligi qanday $(m/s)$?
$Javob$: $a$) _____
$b$) Og‘irlik kuchining bajargan ishi qanday $(J)$?
$Javob$: $b$) _____

84.
Suvdgi nur intensivligi 27 $W/m^2$ ga teng.
$a$) Nurning energiya zichligi qanday$(J/m^3)$?
$Javob$: $a$) _____
$b$) Bu nur suvdagi ko‘zguga qanday bosim beradi $(Pa)$?
$Javob$: $b$) _____

85.
Proton induksiyasi $B$ bo‘lgan magnit maydonga uchib kirdi. Bunda $r$ radiusli aylana chizadi.
$a$) Aylanish chastotasi qanday?
$Javob$: $a$) _____
$b$) Protonning tezligi qanday?
$Javob$: $b$) _____

86.
Quyosh diametri $13{,}8 \cdot 10^8$ $m$ quyosh va yer orasidagi masofa 150 ming $km$. Quyoshning haqiqiy tasviri fokus masofasi 50 $sm$ bo‘lgan linzada hosil qilinmoqda.
$a$) Quyosh tasviri linzadan qanday masaofada bo‘ladi$(mm)$?
$Javob$: $a$) _____

$b$) Quyoshning tasvirdagi diametri necha$(mm)$?
$Javob$: $b$) _____

87.
G‘alayonlangan vodorod atomi ionlanish energiyasi 3,4 $eV$. Foton yutish natijasida atomning impuls momenti 2ħ ga ortdi.
$a$) Keying ionlanish energiyasini toping$(eV)$.
$Javob$: $a$) _____
$b$) Foton yutishi natijasida tezligi qanday o‘zgargan?.
$Javob$: $b$) _____

88.
Balandligi 3 $m$ bo‘lgan 5 $m$ uzunlikdagi qiya tekislikning yuqori uchida turgan jism pastga boshlang‘ich tezliksiz qo‘yib yuborildi va qiyalik oxirida 6 $m/s$ tezlikka erishdi.
$a$) Qiya tekislikni foydali ish koeffitsiyentini toping.
$Javob$: $a$) _____
$b$) Ishqalanish koeffitsiyentini toping.
$Javob$: $b$) _____

89.
Usti ochiq idish 200 $K$ haroratdan 400 $K$ gacha qizdirildi. So‘ng germetik berk qilinib 320 $K$ gacha sovutildi.
$a$) Birinchi holatda gaz bosimi qanday o‘zgargan.
$Javob$: $a$) _____
$b$) Ikkinchi holatda gaz bosimi qanday o‘zgargan.
$Javob$: $b$) _____

90.
Rubob torining uzunligi 0,9 $m$, tarangligi 3 $N$ va massasi esa 3 $g$.
$a$) Tor tebranishining tezligini toping$(m/s)$.
$Javob$: $a$) _____
$b$) Tor tarangligi 4 marta, ortsa tezlik qanday o‘zgaradi?
$Javob$: $b$) _____

91.
O‘zgaruvchan tok zanjiriga qarshliklari mos ravishda 8 Ω, 11 Ω va 5 Ω bo‘lgan aktiv, induktiv va sig‘im qarshliklar ketma-ket ulangan.
$a$) Zanjiradgi quvvat koeffitsiyentini toping.
$Javob$: $a$) _____
$b$) Zanjirning umumiy qarshligini (Ω) toping.
$Javob$: $b$) _____

92.
Birinchi idishda $12 \cdot 10^{30}$ ta, ikkinchi idishda $5 \cdot 10^{30}$ ta atomi bor idishlar o‘zaro tutashtrilgan. Birinchi idishdagi bosim 40 $kPa$ ga ikkinchi idishdagi bosim 25 $kPa$ ga teng.
$a$) Kran ochilsa natijaviy bosimni ($kPa$) toping.
$Javob$: $a$) _____
$b$) Kran ochilsa ikkinchi idishdagi bosim ($kPa$) qanchaga ortadi?
$Javob$: $b$) _____

93.
Massasi $M$ zaryadi $Q$ bo‘lgan qo‘zg‘almas zaryad atrofida, massasi $m$ zaryadi $q$ bo‘lgan zaryadli zarra $r$ radiusli aylanma trayektoriya bo‘ylab harakat qilmoqda. Zaryadlar orasidagi gravitatsiya kuchini hisobga olmang. $M >> m, Q > 0, q > 0$.
$a$) Kichik massali zaryad tezligini toping.
$Javob$: $a$) _____
$b$) Kichik zaryad kinetik energiyasini toping.
$Javob$: $b$) _____

94.
Impulsi $P$, to‘la energiyasi $4Pc$ bo‘lgan zarracha mavjud.
$a$) Bu zarrachani tinchlikdagi massasini toping.
$Javob$: $a$) _____
$b$) Shu zarrani impulsi $2P$ bo‘lgandagi to‘la energiyasini toping.
$Javob$: $b$) _____

95.
Massasi 1 $kg$ tezligi 3 $m/s$ bo‘lgan jism sharqqa tomon, massasi 0,5 $kg$ tezligi 8 $m/s$ bo‘lgan jism janubga harakatlanmoqda.
$a$) Sistemaning massa markazini tezligini ($m/s$) toping.
$Javob$: $a$) _____
$b$) Jismlar noelastik to‘qnashganda qanday energiya ($J$) issiqlikka aylanadi.
$Javob$: $b$) _____

96.
2 $\mu g$ chang zarrasi bilan $12c/13$ tezlikdagi praton impulslari teng.
$a$) Pratonni impulsini ($kg \cdot m/s$) toping.
$Javob$: $a$) _____
$b$) Chang zarrasini tezligini ($m/s$) toping.
$Javob$: $b$) _____

97.
Radiaktiv preparatning 5 minut davomida 15/16 qismi yemrildi.
$a$) Moddaning yarim yemrlish davrini ($s$) toping.
$Javob$: $a$) _____
$b$) Yerim yemrilish doimiysini ($1/s$) toping.

*Javob*: *b*) _____



# JAVOBLAR

| KINEMATIKA | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| A | B | C | A | D | B | C | B | A | A |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| B | B | C | A | B | B | B | B | A | C |
| 21 | 22 | | | | | | | | |
| A | C | | | | | | | | |

| DINAMIKA | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| B | D | A | B | D | D | A | C | B | C |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| B | C | D | B | A | D | C | A | A | A |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | | |
| A | B | D | B | C | A | B | B | | |

| STATIKA | | | | | |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| B | A | B | A | B | D |

| SAQLANISH QONUNLARI | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| D | A | B | D | A | A | B | B | A | C |
| 11 | | | | | | | | | |
| A | | | | | | | | | |

| BOSIM. GIDRO VA AERODINAMIKA | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| C | C | A | B | B | C | C | D | B | A |
| 11 | 12 | | | | | | | | |
| D | C | | | | | | | | |

| MEXANIK TEBRANISHLAR VA TO‘LQINLAR | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| D | C | A | B | A | A | B | D | A | A |
| 11 | 12 | | | | | | | | |
| A | A | | | | | | | | |

| MOLEKULYAR FIZIKA | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| B | D | B | D | C | C | C | A | B | B |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| A | B | A | B | A | A | B | A | C | B |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| A | A | A | A | A | D | C | B | A | A |

| TERMODINAMIKA | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| B | B | B | C | A | B | B | D | A | B |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | | |
| C | A | C | D | A | D | A | B | | |

| ELKTROSTATIKA | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| D | D | A | C | C | C | A | A | D | A |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| B | A | D | A | B | B | B | B | D | C |
| 21 | | | | | | | | | |

| A |
|---|

| O‘ZGARMAS TOK QONUNLARI. TURLI MUHITLARDA ELEKTR TOKI | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| C | C | B | C | D | B | B | B | D | D |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| B | A | A | B | A | C | C | A | D | A |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | |
| C | C | C | C | A | A | C | A | A | |

| ELEKTROMAGNETIZM | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| B | A | B | C | B | D | C | A | D | C |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| D | B | A | C | A | C | A | A | A | B |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| B | B | D | B | B | B | D | C | A | A |
| 31 | 32 | 33 | 34 | | | | | | |
| A | B | B | A | | | | | | |

| OPTIKA | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| C | B | C | A | A | D | D | A | A | B |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| A | C | B | B | A | A | A | A | B | A |
| 21 | 22 | 23 | | | | | | | |
| A | C | A | | | | | | | |

| NISBIYLIK NAZARIYASI VA KVANT FIZIKASI | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| B | D | B | B | B | A | A | B | A | A |
| 11 | 12 | 13 | 14 | 15 | 16 | | | | |
| C | B | A | A | A | A | | | | |

| ATOM VA YADRO FIZIKASI | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| C | A | A | A | A | B | A | A | A | C |
| 11 | 12 | 13 | 14 | | | | | | |
| A | B | A | D | | | | | | |

## MOSLASHTIRISH TALAB QILINADIGAN SAVOLLAR

| | | | |
|---|---|---|---|
| 1 | 33 | 34 | 35 |
| | D | E | B |

| | | | |
|---|---|---|---|
| 2 | 33 | 34 | 35 |
| | E | D | A |

| | | | |
|---|---|---|---|
| 3 | 33 | 34 | 35 |
| | C | F | A |

| | | | |
|---|---|---|---|
| 4 | 33 | 34 | 35 |
| | A | C | E |

| | | | |
|---|---|---|---|
| 5 | 33 | 34 | 35 |
| | A | F | B |

| | | | |
|---|---|---|---|
| 6 | 33 | 34 | 35 |
| | C | A | F |

| | | | |
|---|---|---|---|
| 7 | 33 | 34 | 35 |
| | A | F | B |

| | | | |
|---|---|---|---|
| 8 | 33 | 34 | 35 |
| | D | B | F |

| | | | |
|---|---|---|---|
| 9 | 33 | 34 | 35 |
| | D | E | A |

ISAK

**OCHIQ SAVOLLAR**

| | | | | | |
|---|---|---|---|---|---|
| **1** | ***a*** | 10 | **8** | ***a*** | 25 |
| | ***b*** | 20 | | ***b*** | 1562,5 |
| **2** | ***a*** | 35 | **9** | ***a*** | 360 |
| | ***b*** | 120 | | ***b*** | 1080 |
| **3** | ***a*** | 12 | **10** | ***a*** | 2 |
| | ***b*** | $12\sqrt{3}$ | | ***b*** | 40 |
| **4** | ***a*** | $2kt^2$ | **11** | ***a*** | 12,6 |
| | ***b*** | $k\sqrt{1+\frac{k^2t^2}{r^2}}$ | | ***b*** | 3 |
| **5** | ***a*** | 2 | **12** | ***a*** | 40/3 |
| | ***b*** | 32 | | ***b*** | 6 |
| **6** | ***a*** | 80 | **13** | ***a*** | $\sqrt{g(3h-R)}$ |
| | ***b*** | $-640$ | | ***b*** | $g\cdot\frac{h-R}{R}$ |
| **7** | ***a*** | 0,8 | **14** | ***a*** | 20 |
| | ***b*** | $-800$ | | ***b*** | 0,25 |

| | | | | | |
|---|---|---|---|---|---|
| **15** | ***a*** | $\sqrt{G\frac{M}{r}}$ | **22** | ***a*** | 5,66 |
| | ***b*** | $2\pi\sqrt{\frac{r^3}{GM}}$ | | ***b*** | 80 % |
| **16** | ***a*** | $m_1\vec{\vartheta}_1 + m_2\vec{\vartheta}_2 + m_3\vec{\vartheta}_3$ | **23** | ***a*** | 316 |
| | ***b*** | $\frac{m_1\vec{\vartheta}_1 + m_2\vec{\vartheta}_2 + m_3\vec{\vartheta}_3}{m_1 + m_2 + m_3}$ | | ***b*** | 1,053 |
| **17** | ***a*** | 5 | **24** | ***a*** | 2,5 |
| | ***b*** | 1 | | ***b*** | 4 |
| **18** | ***a*** | $\frac{m_2}{m_1 + m_2}l$ | **25** | ***a*** | $2 \cdot 10^8$ |
| | ***b*** | $\frac{1}{2\pi}\sqrt{\frac{m_1 + m_2}{m_1 m_2} \cdot \frac{1}{k}}$ | | ***b*** | 0,1 % |
| **19** | ***a*** | 5 | **26** | ***a*** | 4 |
| | ***b*** | 1 | | ***b*** | $10^{-21}$ |
| **20** | ***a*** | 6 | **27** | ***a*** | 150 |
| | ***b*** | 4 | | ***b*** | 623 |
| **21** | ***a*** | 88 | **28** | ***a*** | 125 |
| | ***b*** | 8,8 | | ***b*** | 5194 |

| | | | | | |
|---|---|---|---|---|---|
| **29** | ***a*** | 1500 | **36** | ***a*** | $U$ |
| | ***b*** | 4,5 | | ***b*** | $4CU^2$ |
| **30** | ***a*** | 750 | **37** | ***a*** | 83,3 |
| | ***b*** | 300 | | ***b*** | 4000 |
| **31** | ***a*** | $-60$ | **38** | ***a*** | 6,25 |
| | ***b*** | 1,6 | | ***b*** | 75 |
| **32** | ***a*** | 50,4 | **39** | ***a*** | $\sqrt{3}k\frac{q_1q}{a^2}$ |
| | ***b*** | 10 | | ***b*** | $2k\frac{q_1q}{a}$ |
| **33** | ***a*** | $6{,}67 \cdot 10^{23}$ | **40** | ***a*** | $\left(\frac{1}{\sqrt{2}}-3\right)k\frac{q}{l}$ |
| | ***b*** | $10{,}24 \cdot 10^{11}$ | | ***b*** | $2\sqrt{2}k\frac{q}{l^2}$ |
| **34** | ***a*** | 525 | **41** | ***a*** | 525 |
| | ***b*** | 1,3125 | | ***b*** | 1312,5 |
| **35** | ***a*** | $6 \cdot 10^8$ | **42** | ***a*** | $\frac{4\sqrt{3}}{2}$ |
| | ***b*** | 2,88 | | ***b*** | $\frac{4\sqrt{2}}{5}$ |

| | | | | | |
|---|---|---|---|---|---|
| **43** | ***a*** | $4 \cdot 10^4$ | **50** | ***a*** | $\sqrt{17}$ |
| | ***b*** | 7500 | | ***b*** | 2 |
| **44** | ***a*** | $3\sqrt{2}$ | **51** | ***a*** | 2 |
| | ***b*** | 720 | | ***b*** | 12 |
| **45** | ***a*** | 28 | **52** | ***a*** | $\frac{m}{2}$ |
| | ***b*** | 240 | | ***b*** | $\frac{mc^2}{8}$ |
| **46** | ***a*** | 3,57 | **53** | ***a*** | $1{,}8 \cdot 10^8$ |
| | ***b*** | 22,5 | | ***b*** | 25/16 |
| **47** | ***a*** | 16 | **54** | ***a*** | $\frac{2\sqrt{5}}{5}mc$ |
| | ***b*** | 4800 | | ***b*** | $\frac{4\sqrt{10}}{5}$ |
| **48** | ***a*** | 2/3 | **55** | ***a*** | 24 |
| | ***b*** | $7c/10$ | | ***b*** | 40 |
| **49** | ***a*** | +2,5 | **56** | ***a*** | o‘zgarmaydi |
| | ***b*** | 8/7 | | ***b*** | $\frac{25}{16}$ marta ortadi |

| | | | | | |
|---|---|---|---|---|---|
| **57** | ***a*** | $1{,}4c$ | **64** | ***a*** | 87 |
| | ***b*** | $35c/37$ | | ***b*** | 138 |
| **58** | ***a*** | $4E/3$ | **65** | ***a*** | $+1{,}51$ |
| | ***b*** | $16P/9$ | | ***b*** | $+13{,}6$ |
| **59** | ***a*** | $1{,}62 \cdot 10^8$ | **66** | ***a*** | $+1{,}51$ |
| | ***b*** | $2{,}25 \cdot 10^8$ | | ***b*** | $+3{,}4$ |
| **60** | ***a*** | 6 | **67** | ***a*** | 16 marta ortadi |
| | ***b*** | 240 | | ***b*** | 64 marta ortadi |
| **61** | ***a*** | 50 | **68** | ***a*** | 0,25 |
| | ***b*** | 87,5 | | ***b*** | 7 |
| **62** | ***a*** | 12 | **69** | ***a*** | 4 |
| | ***b*** | $A/8$ | | ***b*** | 36 |
| **63** | ***a*** | $1/4$ | **70** | ***a*** | 5 |
| | ***b*** | 1 | | ***b*** | 3,6 |

| | | | | | |
|---|---|---|---|---|---|
| **71** | ***a*** | 4,5 | **78** | ***a*** | 120 |
| | ***b*** | 18 | | ***b*** | 0,6 |
| **72** | ***a*** | 42,85 | **79** | ***a*** | 12 |
| | ***b*** | 9 | | ***b*** | 1600 |
| **73** | ***a*** | 300 | **80** | ***a*** | 3 marta oshgan |
| | ***b*** | 92 | | ***b*** | 24 |
| **74** | ***a*** | 0,4 | **81** | ***a*** | 0 |
| | ***b*** | 30 | | ***b*** | 215 |
| **75** | ***a*** | $7E/12$ | **82** | ***a*** | $\frac{1}{2}Eel$ |
| | ***b*** | $5E/12$ | | ***b*** | 0 |
| **76** | ***a*** | $0{,}2c$ | **83** | ***a*** | 12 |
| | ***b*** | $0{,}38c$ | | ***b*** | 360 |
| **77** | ***a*** | 160 | **84** | ***a*** | $12\cdot 10^{-8}$ |
| | ***b*** | $3{,}5\cdot 10^{-28}$ | | ***b*** | $24\cdot 10^{-8}$ |

| | | | | | |
|---|---|---|---|---|---|
| **85** | ***a*** | $\frac{Ber}{2\pi m}$ | **92** | ***a*** | 34 |
| | ***b*** | $\frac{Ber}{m}$ | | ***b*** | 6 $kPa$ ga kamaydi |
| **86** | ***a*** | 500 | **93** | ***a*** | $\sqrt{k\frac{Qq}{mr}}$ |
| | ***b*** | 4,6 | | ***b*** | $k\frac{Qq}{2r}$ |
| **87** | ***a*** | 0,85 | **94** | ***a*** | $\sqrt{15}\frac{P}{c}$ |
| | ***b*** | 2 marta kamaygan | | ***b*** | $\sqrt{19}Pc$ |
| **88** | ***a*** | 500/7 | **95** | ***a*** | 10/3 |
| | ***b*** | 0,3 | | ***b*** | 12,2 |
| **89** | ***a*** | o‘zgarmagan | **96** | ***a*** | $1{,}2\cdot 10^{-18}$ |
| | ***b*** | 1,25 marta kamaygan | | ***b*** | $6\cdot 10^{-10}$ |
| **90** | ***a*** | 30 | **97** | ***a*** | 75 |
| | ***b*** | 2 marta ortadi | | ***b*** | $9{,}3\cdot 10^{-3}$ |
| **91** | ***a*** | 0,8 | **98** | ***a*** | |
| | ***b*** | 10 | | ***b*** | |


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